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Abstract

An abstract is a short summary of a longer work. The abstract concisely reports the aims and outcomes of your research so that readers know exactly what your paper is about.

Kurzfassung

Eine Kurzfassung ist eine kurze Zusammenfassung deiner Arbeit. Die Kurzfassung berichtet prägnant über die Ziele und Ergebnisse deiner Forschung, sodass Leser genau wissen worum es in der Arbeit geht.

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Notation

Table 1: Modellierung Symbole

Symbol	Code	Beschreibung
φ_e, φ_m		Rotorposition
ω_e, ω_m	`we, wm`	Rotorgeschwindigkeit elektrisch, mechanisch
ψ		magnetische Flussverkettung
u		elektrische Spannung - immer die Stellgröße im Regelkreis
i		
$i_{S,filter}$		aufgrund Strommesskette gefilterter Statorstrom
p		Polpaarzahl
L		
R		

Table 2: Regelung Symbole

Symbol	Code	Beschreibung
A		Systemmatrix $\mathbb{R}^{n\times n}$
B		Eingangsmatrix $\mathbb{R}^{m\times n}$
C		Ausgangangsmatrix $\mathbb{R}^{n\times q}$
E		Affiner Term des Zustandsmodells, induzierte Spannung in RRF
$\boldsymbol{x}$		Zustandsvektor $\mathbb{R}^{n\times 1}$
ξ		Erweiterter Zustandsvektor
u		Eingangsvektor $\mathbb{R}^{m\times 1}$
$\Delta u, \bar{\Delta} u$	`Delta_uk(_b)`	Differentieller Eingangsvektor, $\overline{\text{Prädiktionsvektor}}$ für Δu über N_P
y		Ausgang $\mathbb{R}^{q\times 1}$
N_P, N_C, D_S		Prädiktionshorizont, Regelungshorizont, Entscheidungsschritte
d		Störung
$P_x, P_u, P_{\bar{\Delta} u}, P_{E\omega}$		Vorhersage-Matrizen respektive $x, u, \bar{\Delta} u$, $E(\omega[k])$
T_S		Abtastzeit, Regelkreis
Λ, λ		Luenberger Beobachter Verstärkung
η		Soft Constraint
$\boldsymbol{\rho}$		Skalierungs/Aktivierungsvektor für skalares η

Table 3: Symboldekorationen mit x als Beispielsymbol

Symbol	Code	Beschreibung
$\boldsymbol{x}, \boldsymbol{X}$		Vektor, Matrix
$\dot{x}, \ddot{x}$	Präfix d	1., 2. zeitliche Ableitung
$\hat{x}$	Suffix h	geschätzte Größe
$\overline{\boldsymbol{x}}, \overline{\boldsymbol{X}}$	Suffix b	prädizierter Vektor (über N_P) , für Prädiktion benötigte Matrix
$\tilde{x}$	Suffix t	Modifikation, Abwandlung von x
$\underline{x}$		komplexer Zeiger
x^*	Suffix opt	optimale Größe

Table 4: Koordinatensysteme

Symbol	Code	Beschreibung
α, β	SRF	Stator festes KS
d, q	RRF	Rotorfestes KS
a, b, c		Motoranschluss basiert
$\boldsymbol{T}_{abc\to\alpha\beta}$	`T.C`	Transformation from 3 phase system to SRF
$\boldsymbol{T}_{\alpha\beta\to dq}(\varphi[k])$		Transformation form SRF to RRF for current $\varphi[k]$
$\bar{\boldsymbol{T}}_{\alpha\beta\to dq}(\bar{\varphi}[k])$		$\boldsymbol{T}_{\alpha\beta\to dq}(\varphi[k])$ for prediction vector
$\boldsymbol{T}_{\alpha\beta\to abc} = \boldsymbol{T}^{\dagger}_{abc\to\alpha\beta}$	`T.iC`	Inverse Transform: SRF to 3 phase
$\boldsymbol{T}_{dq\to\alpha\beta}(\varphi[k]) = \boldsymbol{T}_{\alpha\beta\to dq}(\varphi[k])^{-1}$		Inverse Transform: RRF to SRF

Table 5: Operatoren

Symbol	Code	Beschreibung
$\sigma()$		Spektrum einer Matrix
s		Laplace Variable
z		Z-Transformation Variable
$\boldsymbol{X}^{\dagger}$		Moore-Penrose Pseudo-Inverse
$\|\boldsymbol{x}\|^2_{\boldsymbol{Q}}$		Quadrtic Form of $\boldsymbol{x}$ with weighting $\boldsymbol{x}^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{x}$
$\nabla_{\boldsymbol{x}} J$		Gradient of J with respect to x
$\nabla^2_{\boldsymbol{x}} J$		Hessian Matrix J with respect to x
$\Sigma :$		Systemdefinition (Zustandsraummodell)

1

Introduction

1.1 Space Vector Theory

Due to benefits regarding the mathematical modeling of electrical three-phase system, space vectors are commonly used. The idea behind it, is to describe real-valued three-phase quantities as one complex-valued quantity. By doing so, a change of reference frames is easily achieved (as shown in Section 1.2) and an intuitive description of the magnitude and phase angle of a three-phase system is provided.

For a balanced sinusoidal three-phase current system, the phase currents can be expressed as

$$
\begin{aligned}
i_a(t) &= I \cos \omega t, \\
i_b(t) &= I \cos\left(\omega t - \frac{2\pi}{3}\right), \\
i_c(t) &= I \cos\left(\omega t + \frac{2\pi}{3}\right)
\end{aligned}
$$

Those sinusoidal waves with constant angular frequency can be expressed as complex quantities $\hat{I}$, $\hat{I}e^{\mathrm{j}2\pi/3}$, $\hat{I}e^{-\mathrm{j}2\pi/3}$, with fixed angle of 120° relative to each other, as depicted in Figure 1.1.

The instantaneous phase quantities can be represented by the complex space vector

$$
\underline{i}_S = \frac{2}{3}\left(I + Ie^{\mathrm{j}2\pi/3} + Ie^{-\mathrm{j}2\pi/3}\right).
$$

The scaling factor 2/3 is chosen according to the normalization convention such that the magnitude of the space vector corresponds to the amplitude of the balanced three-phase quantity. In general, the use of a complex space vector does not require the three-phase quantities to be sinusoidal or to have a constant frequency Defining $i_\alpha = \mathfrak{Re}\{\underline{i}_{abc}\}$ and $i_\beta = \mathfrak{Im}\{\underline{i}_{abc}\}$, the commonly known Clarke transform for balanced systems, as in three phase system without connection of the neutral star point, can be derived.

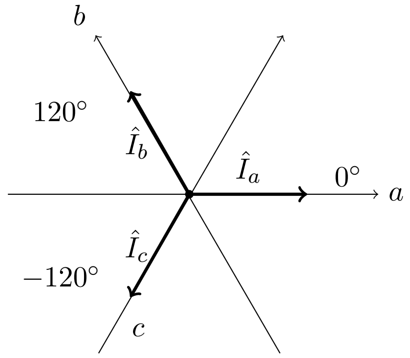


Figure 1.1: Three phase system definition with complex quantities

Even though this transformation seems like a projection, no information is lost when the zero sequence condition $i_a + i_b + i_c = 0$ is considered.

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \underbrace{\frac{2}{3} \begin{bmatrix} 1 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{\sqrt{3}}{2} & -\frac{\sqrt{3}}{2} \end{bmatrix}}_{\boldsymbol{T}_{abc\rightarrow\alpha\beta}} \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}. \tag{1.1}$$

Even though $\boldsymbol{T}_{abc\rightarrow\alpha\beta}$ is not a square matrix, the inverse transform can be defined via the Moore-Penrose pseudo-inverse $\boldsymbol{T}_{\alpha\beta\rightarrow abc} = \boldsymbol{T}^{\dagger}_{abc\rightarrow\alpha\beta}$.
The Clarke transform can be thought of as a convenient mathematical description for the three phase quantities by using a space vector. As mentioned, this eases the transformation between coordinate systems. This will be shown in the following section.

1.2 Reference Frames

Two reference frame for the stator and voltage current space vector will be of interest. The first one is the stationary reference frame (SRF). In this frame of reference the space vector is a direct representation of the quantities, which can be directly measured at the terminals of the machine. Therefore the constraints, due to the generation of the stator voltage is stationary in this frame of reference. Due to the orientation of the magnetic main flux linkage relative to the rotor, it is beneficial to also describe these quantities in a reference frame fixed to the rotor. This frame will be called the rotating reference frame (RRF). By doing so, the real or direct component of this vector $\mathfrak{Re}\{\underline{i}_S^R\} = i_d$ is oriented in the same axis as the flux linkage of the permanent magnets $\underline{\psi}_{PM}$ and the imaginary or

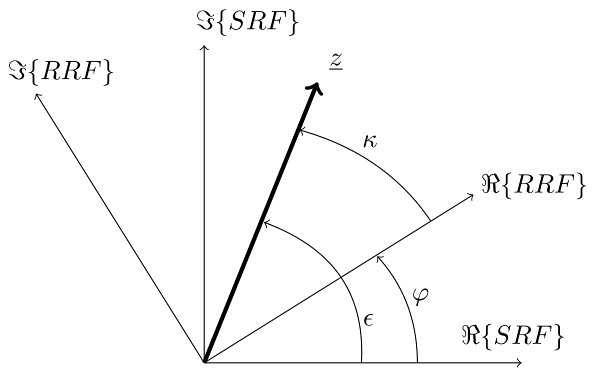


Figure 1.2: Arbitrary space vector $\underline{z}$ in SRF and RRF

quadrature component $\mathfrak{Im}\{\underline{i}_S^R\} = i_q$ will be rotated by 90°.Consequently, the D axis current i_d primarily influences the flux-producing component of the stator current, whereas the Q axis current i_q can be interpreted as the torque-producing component of the stator current.

1.2.1 Coordinate Frame Transformation

As mentioned in the previous section, a transformation rule is needed, to translate quantities from the SRF to the RRF and vice versa.

To describe this transformation an arbitrary space vector $\underline{z}$ is considered. In the stationary reference frame, the vector can be expressed as

$$\underline{z}^S = z_\alpha + \mathrm{j}z_\beta = |z|e^{\mathrm{j}\epsilon},$$

where ϵ denotes the angle of the space vector relative to the α-axis. Therefore, its components are given by

$$z_\alpha = \mathfrak{Re}\{\underline{z}^S\} = |z|\cos\epsilon, \quad z_\beta = \mathfrak{Im}\{\underline{z}^S\} = |z|\sin\epsilon.$$

The same space vector can be expressed in the rotating reference frame as

$$\underline{z}^R = z_d + \mathrm{j}z_q = |z|e^{\mathrm{j}\kappa},$$

where κ denotes the angle of the space vector relative to the D axis. The corresponding components are then give as

$$z_d = \mathfrak{Re}\{\underline{z}^R\} = |z|\cos\kappa, \quad z_q = \mathfrak{Im}\{\underline{z}^R\} = |z|\sin\kappa.$$

The angle between the α-axis of the SRF and the d-axis of the RRF is defined by the electrical angle $\varphi_e = \varphi$, as illustrated in Figure 1.2. Due to that displacement, the space vector is transformed from the stationary to the rotating reference frame by a rotation of $-\varphi$. Hence,

$$\underline{z}^R = \underline{z}^S e^{-\mathrm{j}\varphi}$$

holds.
Substituting the Cartesian representation of the space vector gives

$$z_d + \mathrm{j}z_q = (z_\alpha + \mathrm{j}z_\beta)(\cos\varphi - \mathrm{j}\sin\varphi). \tag{1.2}$$

Expanding the right-hand side and comparing the real and imaginary parts yields

$$\begin{aligned} z_d &= z_\alpha \cos\varphi + z_\beta \sin\varphi, \\ z_q &= -z_\alpha \sin\varphi + z_\beta \cos\varphi. \end{aligned}$$

The transformation can consequently be written in matrix form as

$$\begin{bmatrix} z_\alpha \\ z_\beta \end{bmatrix} = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} z_d \\ z_q \end{bmatrix}. \tag{1.3}$$

Since this transformation rule can be applied for a any vector quantity, the stator current can be transformed between the two reference frames according to

$$\begin{bmatrix} i_d \\ i_q \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}}_{\boldsymbol{T}_{\alpha\beta\to dq}(\varphi)} \begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix}.$$

The inverse transformation is then given by

$$\begin{bmatrix} i_\alpha \\ i_\beta \end{bmatrix} = \underbrace{\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}}_{\boldsymbol{T}_{dq\alpha\beta}(\varphi)} \begin{bmatrix} i_d \\ i_q \end{bmatrix}.$$

Since $\boldsymbol{T}_{\alpha\beta\rightarrow dq}(\varphi)$ is a orthonormal matrix, $\boldsymbol{T}_{dq\alpha\beta}(\varphi) = (\boldsymbol{T}_{\alpha\beta\rightarrow dq}(\varphi))^{-1} = (\boldsymbol{T}_{\alpha\beta\rightarrow dq}(\varphi))^{\mathrm{T}}$ holds [11].

1.3 2-Level Voltage Source Inverter

To generate the phase voltages for the control of the machine, while maintaining the high power required to operate it, a switched system is needed. This is commonly achieved using a two-level voltage source inverter (2L-VSI). The topology of such an 2L-VSI is depicted in Figure 1.3. It consists of one half bridge for each of the three phases, supplied by an DC voltage source (or rectified AC voltage).

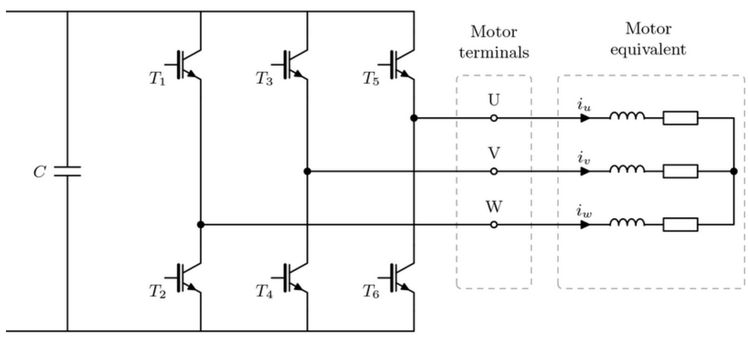


Figure 1.3: Two-level inverter driving a 3-phase motor. Image source: [15]

In theory, each of the six semiconductor switches can be controlled independently. However, during normal operation, these switches are operated in a complementary manner. Turning on both switches simultaneously would short-circuit the DC link and cause a shoot-through event. To prevent this, a short dead time is typically introduced when changing between switching states, during which both switches of a half-bridge are turned off. During this interval, the stator current can continue to flow through the freewheeling diodes.

Since the PMSM considered here has no neutral connection, it will only react on the differential voltage signals with respect to the half bridge output nodes. Apart from the dead-time during interlocking, the state of the upper switch directly determines the state of the lower switch. With that, there $2^3 = 8$ possible states in which the switches can be controlled. By defining $\boldsymbol{s}_{upper}^{\mathrm{T}} = [s_{a,u}\ s_{b,u}\ s_{c,u}]$ as the vector of the state of the upper switches, the set of all switching states can be written down:

$$\mathcal{S}_{upper} = \left\{ \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}, \begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix} \right\}$$

The first and last vector of $\mathcal{S}_{upper}$ result in identical voltages at all three half-bridge output nodes and therefore result in a zero stator voltage space vector. Thus, they are referred to as the zero voltage vectors. The remaining six active vector can be used to direct the current flow through the machine.
By applying the Clarke transform to the a vector of switching states, scaled by the DC-Link voltage of the inverter U_{DC}, the corresponding voltage space vectors can be determined.

$$\begin{bmatrix}u_\alpha\\u_\beta\end{bmatrix} = U_{DC} \cdot \boldsymbol{T}_{abc\to\alpha\beta} \cdot \begin{bmatrix}s_{a,u}\\s_{b,u}\\s_{c,u}\end{bmatrix}$$

Applying this transformation to all eight switching states results in six equally spaced active voltage vectors and two zero voltage vectors located at the origin. The six active vectors form the well-known voltage hexagon of a 2L-VSI, as depicted in Figure 1.4. This voltage hexagon defines the set of voltage space vectors that can be directly generated by the inverter

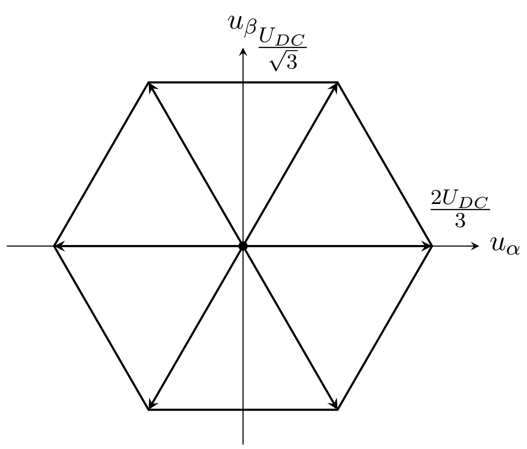


Figure 1.4: Voltage Hexagon in SRF

1.4 Model Predictive Control

1.4.1 Comparison to classical control

Using classical field-oriented PI current control, this is solved by splitting the system up into two linear, decoupled subsystems and two coupling terms (with the induced voltage acting affine to one of the).

$$\begin{aligned}
\dot{i}_{Sd} &= \underbrace{\frac{1}{L_d} u_{Sd} - \frac{R_S}{L_d} i_{Sd}}_{\Sigma_{\mathrm{d,decoupled}}} + \underbrace{\frac{L_q}{L_d} i_{Sq}\omega}_{\Sigma_{\mathrm{d,coupled}}} \\
\dot{i}_{Sq} &= \underbrace{\frac{1}{L_q} u_{Sq} - \frac{R_S}{L_q} i_{Sq}}_{\Sigma_{\mathrm{q,decoupled}}} - \underbrace{\frac{L_d}{L_q} i_{Sd}\omega}_{\Sigma_{\mathrm{q,coupled}}} - \underbrace{\frac{1}{L_q} \psi_{PM}\omega}_{\Sigma_{\mathrm{q,affine}}}
\end{aligned}$$

The linear systems $\Sigma_{\mathrm{d,decoupled}}$ and $\Sigma_{\mathrm{q,decoupled}}$ are controlled using two independent PI controllers. The remaining terms are compensated by a feed-forward system.

2

Modelling

2.1 Interior Magnet PMSM

$$\underline{u}_S^S = R_S \underline{i}_S^S + \dot{\underline{\psi}}_S^S \tag{2.1}$$

$$\underline{u}_S^R = R_S \underline{i}_S^R + \dot{\underline{\psi}}_S^R + \mathrm{j}\omega_e \underline{\psi}_S^R \tag{2.2}$$

... This leads to the to the continuous time state space model of Equation (2.3).

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix} = \begin{bmatrix} -\frac{R_S}{L_d} & \frac{L_q}{L_d}\omega_e(t) \\ -\frac{L_d}{L_q}\omega_e(t) & -\frac{R_S}{L_q} \end{bmatrix} \begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix} \begin{bmatrix} v_{Sd} \\ v_{Sq} \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{\psi_{PM}}{L_q}\omega_e(t) \end{bmatrix} \tag{2.3}$$

To analyze the Model (2.3) in the context of linear parameter-varying systems, the permanent magnet flux linkage ψ_{PM} is sometimes treated as a state or input [?]. This results in the following description, which matches the generic LPV system form as defined in [1] $\dot{\boldsymbol{x}} = \boldsymbol{A}(\omega)\boldsymbol{x} + \boldsymbol{B}(\omega)$.

$$\frac{d}{dt}\begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix} = \begin{bmatrix} -\frac{R_S}{L_d} & \frac{L_q}{L_d}\omega_e(t) \\ -\frac{L_d}{L_q}\omega_e(t) & -\frac{R_S}{L_q} \end{bmatrix} \begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix} + \begin{bmatrix} \frac{1}{L_d} & 0 & 0 \\ 0 & \frac{1}{L_q} & -\frac{\omega_e(t)}{L_q} \end{bmatrix} \begin{bmatrix} v_{Sd} \\ v_{Sq} \\ \psi_{PM} \end{bmatrix} \tag{2.4}$$

Analyzing this model, one has to quantify the influence of the parameter ω. Due to the balance of angular momentum of the motor shaft (2.5), the electrical rotor speed $\omega = \omega_e = \omega_{m1} p$ is Lipschitz-continuous. With the eigenvalues of the respective dynamical system (2.5) being slower than (3.1), as discussed in Section 2.2.

2.2 Coupling Dynamics

$$
\begin{aligned}
J_1 \frac{\mathrm{d}(\omega_{m1})}{\mathrm{d}t} &= T_e - c \cdot \Delta\varphi - k \cdot \Delta\omega_m \\
J_2 \frac{\mathrm{d}(\omega_{m2})}{\mathrm{d}t} &= T_l + c \cdot \Delta\varphi + k \cdot \Delta\omega_m \\
\frac{\mathrm{d}(\varphi_{m1})}{\mathrm{d}t} &= \omega_{m1} \\
\frac{\mathrm{d}(\varphi_{m2})}{\mathrm{d}t} &= \omega_{m2}
\end{aligned}
\tag{2.5}
$$

3

Model Predictive Current Control

3.1 Prediction Model

This section discusses the derivation of the nonlinear, discrete-time prediction of the PMSM's dynamics. First, the decision-making process for this prediction model is outlined in the context of field-oriented current control for PMSMs. Next, the process of discretizing the continuous-time system is explained. For the current controller in this chapter, this is relatively simple; however, it poses greater challenges in the observer-based current prediction controller in Chapter 4. Finally, as is typical for MPC, the prediction is extended to the entire prediction horizon. The assumptions relevant to the prediction are also presented there.

3.1.1 Choice of adequate State Space Model

As derived in Section 2.1, the model of the interior magnet PMSM is given by Equation (3.1).

$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix}}_{\dot{\boldsymbol{x}}_c} = \underbrace{\begin{bmatrix} -\frac{R_S}{L_d} & \frac{L_q}{L_d}\omega(t) \\ -\frac{L_d}{L_q}\omega(t) & -\frac{R_S}{L_q} \end{bmatrix}}_{\boldsymbol{A}_c(\omega(t))} \underbrace{\begin{bmatrix} i_{Sd} \\ i_{Sq} \end{bmatrix}}_{\boldsymbol{x}_c} + \underbrace{\begin{bmatrix} \frac{1}{L_d} & 0 \\ 0 & \frac{1}{L_q} \end{bmatrix}}_{\boldsymbol{B}_c} \underbrace{\begin{bmatrix} u_{Sd} \\ u_{Sq} \end{bmatrix}}_{\boldsymbol{u}_c} + \underbrace{\begin{bmatrix} 0 \\ -\frac{\psi_{PM}}{L_q}\omega(t) \end{bmatrix}}_{\boldsymbol{E}_c(\omega(t))} \tag{3.1}$$

For the ease of notation, $\omega = \omega_e = \omega_{m1}p$ refers to the electrical rotor speed for the rest of this thesis. Writing the nonlinear continuous time dynamic system in established control theory [2] [3] notation leads to the following :

$$\begin{aligned} \dot{\boldsymbol{x}}_c &= \boldsymbol{f}(\boldsymbol{x}_c, \boldsymbol{u}_c, \omega) \\ \dot{\boldsymbol{x}}_c &= \boldsymbol{A}_c(\omega(t))\boldsymbol{x}_c + \boldsymbol{B}_c\boldsymbol{u}_c + \boldsymbol{E}_c(\omega(t)) \end{aligned} \tag{3.2}$$

For a variety of applications it is beneficial to work with linear models, since this allows for the use of broad framework of control schemes. However, the affine term regarding the induced voltage $\boldsymbol{E}_c(\omega(t))$ is non-negligable for the current dynamics $\dot{\boldsymbol{x}}$ at every operating point except standstill or locked rotor operation.

As mentioned in Section $refsec: intro_comp_cs$, using model predictive control, one can directly use a MIMO formulation, leading to a direct consideration the D/Q axis coupling. The question remaining is whether a discretized model of Equation (3.1) shall be directly used or a step-wise linearization approach around an operating point shall be employed. For the linearization, one would define an operating point as

$$\boldsymbol{x}_c = \boldsymbol{x}_{c,\mathrm{OP}} + \boldsymbol{\delta x}_c, \qquad \boldsymbol{u}_c = \boldsymbol{u}_{c,\mathrm{OP}} + \boldsymbol{\delta u}_c, \qquad \omega = \omega_{\mathrm{OP}} + \delta\omega.$$

And approximate the dynamics (3.2) using a first-order Taylor approximation around an operating point $\mathrm{OP} = (\boldsymbol{x}_{c,\mathrm{OP}}, \boldsymbol{u}_{c,\mathrm{OP}}, \omega_{\mathrm{OP}})$

$$\frac{\mathrm{d}\boldsymbol{\delta x}_c}{\mathrm{d}t} = \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}_c}\right|_{\mathrm{OP}} \boldsymbol{\delta x}_c + \left.\frac{\partial \boldsymbol{f}}{\partial \boldsymbol{u}_c}\right|_{\mathrm{OP}} \boldsymbol{\delta u}_c + \left.\frac{\partial \boldsymbol{f}}{\partial \omega}\right|_{\mathrm{OP}} \delta\omega.$$

Analyzing the nonlinear system (3.1), it can be seen that th only non-linearity stems from the induced voltage $\boldsymbol{E}_c(\omega(t))$, which is proportional to the scheduling parameter ω. As shown in Section 2, the dynamics of the mechanical system are slower than the electrical dynamics, assuming $\omega(t)$ is slowly varying relative to the sampling period T_S. Therefore, introducing step-wise linearization only leads to increased complexity without reducing the computational effort.

Thus, it was decided to use the affine prediction model of Equation (3.1) directly. This approach is also typically used in modern MPC PMSM literature [4] [5] [6].

3.1.2 Discretization

Due to its simplicity, Forward Euler discretization shall be considered. This approach conserves stability, when the sampling time T_S is sufficiently small. The mapping of continuous time eigenvalues to discrete time eigenvalues $\underline{s}_i = a + \mathrm{j}b \ : \ a, b \in \mathbb{R} \rightarrow \underline{z}_i$ is described as

$$\underline{z}_i = 1 + T_S \underline{s}_i$$

For $\underline{z}_i$ to be within the unit circle

$$\|\underline{z}_i\| = \|1 + T_S \underline{s}_i\| = 1 + 2T_S^2 a + T_S a^2 + T_S b^2 < 1 \iff T_S^2(a^2 + b^2) < -2T_S a$$

must hold. So, for positive T_S, this can be rewritten as

$$T_S \|\underline{s}_i\|^2 < -2\mathfrak{Re}\{\underline{s}_i\} \iff T_S < \frac{-2\mathfrak{Re}\{\underline{s}_i\}}{\|\underline{s}_i\|^2}$$

With the eigenvalue analysis of Section 2 and $T_S \leq 200\,\mu\text{s}$ within the scope of this thesis, applying the Forward Euler approach will lead to stable discretization for the model of the PMSM. The discretized state space model is then obtained by:

$$\begin{aligned}
&\boldsymbol{A}(\omega[k]) = \boldsymbol{I} + T_S \boldsymbol{A}_c(\omega(kT_S)) \quad \boldsymbol{B} = T_S \boldsymbol{B}_c \\
&\boldsymbol{E}(\omega[k]) = \boldsymbol{E}_c(\omega(kT_S))
\end{aligned}$$

In literature the actuating signal is often split up as $\boldsymbol{u}[k] = \boldsymbol{u}[k-1] + \boldsymbol{\Delta u}[k]$. This mainly offers benefits as it eases the stability analysis [7] [3]. So, the discrete time-model for prediction is given by Equation (3.3).

$$\begin{aligned}
\boldsymbol{x}[k+1] &= \boldsymbol{A}(\omega[k])\boldsymbol{x}[k] + \boldsymbol{B}(\boldsymbol{u}[k-1] + \boldsymbol{\Delta u}[k]) + \boldsymbol{E}(\omega[k]) \\
\boldsymbol{x}[k+1] &= \begin{bmatrix} 1 - \frac{T_s R_S}{L_d} & T_s \frac{L_q}{L_d}\omega[k] \\ -T_s \frac{L_d}{L_q}\omega[k] & 1 - \frac{T_s R_S}{L_q} \end{bmatrix} \boldsymbol{x}[k] \\
&\quad + \begin{bmatrix} \frac{T_s}{L_d} & 0 \\ 0 & \frac{T_s}{L_q} \end{bmatrix} (\boldsymbol{u}[k-1] + \boldsymbol{\Delta u}[k]) + \begin{bmatrix} 0 \\ -\frac{T_s \psi_{PM}}{L_q}\omega[k] \end{bmatrix}
\end{aligned} \tag{3.3}$$

3.1.3 Prediction Horizon

In model predictive control one distinguishes between the prediction horizon N_P, which describes how many discrete steps in time the dynamics are predicted and the control horizon N_C, which is the amount of optimal actuating signal are computed at every step (w)ith $N_C \leq N_P$). Within the scope of this thesis, typically field-oriented current control sampling times of $T_S \in [5, 20]$ kHz are considered. To get an optimal trade off between model fault tolerance, tracking performance and computational effort a typical prediction horizon of $N_P \leq 3$ with $N_P = N_C$ are considered.
To obtain a prediction for $\boldsymbol{x}[k+l]$ $l \in \mathbb{N} \, : \, l \leq N_P$, Equation (3.3) is iterated one step ahead.

$$\hat{\boldsymbol{x}}[k+2] = \boldsymbol{A}(\hat{\omega}[k+1])\hat{\boldsymbol{x}}[k+1] + \boldsymbol{B}(\boldsymbol{u}[k] + \boldsymbol{\Delta u}[k+1]) + \boldsymbol{E}(\hat{\omega}[k+1])$$

Using Relation (3.3) for the estimation $\hat{\boldsymbol{x}}[k+1]$, $\boldsymbol{u}[k] = \boldsymbol{u}[k-1] + \boldsymbol{\Delta u}[k]$ and assuming constant rotor speed for all prediction steps $\hat{\omega}[k+l] = \omega[k]$, leads to

$$
\begin{aligned}
\hat{\boldsymbol{x}}[k+2] &= \boldsymbol{A}(\omega[k])\left(\boldsymbol{A}(\omega[k])\boldsymbol{x}[k] + \boldsymbol{B}(\boldsymbol{u}[k-1] + \boldsymbol{\Delta u}[k] + \boldsymbol{E}(\omega[k])\right)\\
&\quad + \boldsymbol{B}(\boldsymbol{u}[k-1] + \boldsymbol{\Delta u}[k] + \boldsymbol{\Delta u}[k+1]) + \boldsymbol{E}(\omega[k])\\
\hat{\boldsymbol{x}}[k+2] &= (\boldsymbol{A}(\omega[k]))^2 + (\boldsymbol{A}(\omega[k]) + \boldsymbol{I})\,\boldsymbol{B}\boldsymbol{u}[k-1]\\
&\quad + \boldsymbol{B}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{\Delta u}[k] + \boldsymbol{B}\boldsymbol{\Delta u}[k+1] + \left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{E}(\omega[k])
\end{aligned}
$$

This continues up to $l = N_P$.

$$
\begin{aligned}
\hat{\boldsymbol{x}}[k+N_P] &= (\boldsymbol{A}(\omega[k]))^{N_P}\,\boldsymbol{x}[k] + \left(\sum_{f=0}^{N_P-1} (\boldsymbol{A}(\omega[k]))^f\right)\boldsymbol{B}\boldsymbol{u}[k-1]\\
&\quad + \sum_{h=0}^{N_P-1}\left(\sum_{f=0}^{N_P-1-h} (\boldsymbol{A}(\omega[k]))^f\right)\boldsymbol{B}\boldsymbol{\Delta u}[k+h]\\
&\quad + \left(\sum_{f=0}^{N_P-1} (\boldsymbol{A}(\omega[k]))^f\right)\boldsymbol{E}(\omega[k])
\end{aligned}
$$

For the nominal system (3.3), the output is equal to the current states $y[k] = x[k]$, but for other formulations with $\boldsymbol{C} \neq \boldsymbol{I}$ it is necessary to multiply the state description from the left by $\boldsymbol{C}$ to obtain an estimation for the output $\hat{\boldsymbol{y}}[k+l]$. Doing that and using vector notations for all $l \in [1, N_P]$ of $\hat{\boldsymbol{y}}[k+l]$, leads to

$$
\begin{aligned}
\underbrace{\begin{bmatrix} \hat{\boldsymbol{y}}[k+1] \\ \hat{\boldsymbol{y}}[k+2] \\ \vdots \\ \hat{\boldsymbol{y}}[k+N_P] \end{bmatrix}}_{\bar{\boldsymbol{y}}[k+1]} = \underbrace{\begin{bmatrix} \boldsymbol{C}\boldsymbol{A}(\omega[k]) \\ \boldsymbol{C}\left(\boldsymbol{A}(\omega[k])\right)^2 \\ \vdots \\ \boldsymbol{C}\left(\boldsymbol{A}(\omega[k])\right)^{N_P-1} \end{bmatrix}}_{\boldsymbol{P}_x} \boldsymbol{x}[k] + \underbrace{\begin{bmatrix} \boldsymbol{C}\boldsymbol{B} \\ \boldsymbol{C}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{B} \\ \vdots \\ \boldsymbol{C}\left(\sum_{l=0}^{N_P-1}\left(\boldsymbol{A}(\omega[k])\right)^l\right)\boldsymbol{B} \end{bmatrix}}_{\boldsymbol{P}_u} \boldsymbol{u}[k-1] + \\
\underbrace{\begin{bmatrix} \boldsymbol{C}\boldsymbol{B} & \boldsymbol{0}_2 & \cdots & \boldsymbol{0}_2 \\ \boldsymbol{C}(\boldsymbol{A}(\omega[k]) + \boldsymbol{I})\boldsymbol{B} & \boldsymbol{C}\boldsymbol{B} & \cdots & \boldsymbol{0}_2 \\ \boldsymbol{C}\left((\boldsymbol{A}(\omega[k]))^2 + \boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{B} & \boldsymbol{C}(\boldsymbol{A}(\omega[k]) + \boldsymbol{I})\boldsymbol{B} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \boldsymbol{0}_2 \\ \boldsymbol{C}\left(\sum\limits_{l=0}^{N_P-1}(\boldsymbol{A}(\omega[k]))^l\right)\boldsymbol{B} & \boldsymbol{C}\left(\sum\limits_{l=0}^{N_P-2}(\boldsymbol{A}(\omega[k]))^l\right)\boldsymbol{B} & \cdots & \boldsymbol{C}\boldsymbol{B} \end{bmatrix}}_{\boldsymbol{P}_{\bar{\Delta u}}} \underbrace{\begin{bmatrix} \boldsymbol{\Delta u}[k] \\ \boldsymbol{\Delta u}[k+1] \\ \vdots \\ \vdots \\ \boldsymbol{\Delta u}[k+N_P-1] \end{bmatrix}}_{\bar{\boldsymbol{\Delta u}}[k]} \\
+ \underbrace{\begin{bmatrix} \boldsymbol{C} \\ \boldsymbol{C}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right) \\ \vdots \\ \boldsymbol{C}\left(\sum_{l=0}^{N_P-1}\left(\boldsymbol{A}(\omega[k])\right)^l\right) \end{bmatrix}}_{\boldsymbol{P}_E} \boldsymbol{E}(\omega[k])
\end{aligned}
\tag{3.4}
$$

$$
\bar{\boldsymbol{y}}[k+1] = \underbrace{\boldsymbol{P}_x\bar{\boldsymbol{x}}[k] + \boldsymbol{P}_u\boldsymbol{u}[k-1] + \boldsymbol{P}_E\boldsymbol{E}(\omega[k])}_{=:\bar{\boldsymbol{g}}[k]} + \boldsymbol{P}_{\bar{\Delta u}}\bar{\boldsymbol{\Delta u}}[k] \tag{3.5}
$$

With $\bar{\boldsymbol{g}}[k]$ being parameter variant, but independent of $\bar{\boldsymbol{\Delta u}}[k]$. For the implementation it is beneficial to recognize that the columns of $\boldsymbol{P}_{\bar{\Delta u}}$ are down-shifted columns of $\boldsymbol{P}_u$.

3.2 Optimization Problem

The vector of the voltage increment at the current step in time and its predictions up to the N_Pth step shall be determined optimal with respect to a certain cost function J and constraints regarding the stator voltage and the predicted stator current response, such that the determined stator voltage can be physically applied without damaging the machine. For the actuating signal $\bar{\boldsymbol{u}}_S[k]$, this means it has to be bounded by the voltage hexagon and the magnitude of each $\bar{\boldsymbol{i}}_S[k+1]$ must not exceed the current limit I_{max}. Doing so, leads to a general description for MPC for PMSM, similar to the one used in [4].

$$
\begin{aligned}
\bar{\boldsymbol{\Delta u}}_S[k]^* = &\arg\min_{\bar{\boldsymbol{\Delta u}}_S} \; J(\boldsymbol{u}_S[k], \boldsymbol{i}_S[k], \boldsymbol{i}_{S,ref}[k+1]) \\
\text{s.t.} \quad & \hat{\boldsymbol{i}}_S[k+1] \in \mathbb{I}_S = \{\boldsymbol{i}_S \in \mathbb{R}^2 : \|\boldsymbol{i}_S^R[k+1]\|_2 \leq I_{max}\} \\
& \boldsymbol{u}_S^S[k] \in \mathbb{U}_S^S = \{\boldsymbol{u}_S^S \in \mathbb{R}^2 : \boldsymbol{u}_S^S = U_{DC}\boldsymbol{T}_{abc\rightarrow dq}\boldsymbol{d}_{abc}\}
\end{aligned}
$$

Since the controller operates within the RRF, the relation for the voltage constraints must be transformed. To gurantee a unique global minimizer, J must be a convex function and $\mathbb{U}_S^R, \mathbb{I}_S^R$ convex sets, expressed as a set of linear inequality constraints. To use the framework of quadratic programming, the constraints must be expressed in terms of the optimization variable $\bar{\boldsymbol{\Delta u}}[k]$. Keeping this in mind and using index u for quantities regarding the prediction vector of the actuating signal $\bar{\boldsymbol{u}}_S[k]$ and i for quantities regarding the output prediction vector $\bar{\boldsymbol{i}}_S[k+1]$, the goal of this section is to derive a quadratic program in the form of

$$
\begin{aligned}
\bar{\boldsymbol{\Delta u}}_S^R[k]^* &= \arg\min_{\bar{\boldsymbol{\Delta u}}_S} \; \frac{1}{2}\left(\bar{\boldsymbol{\Delta u}}_S^R[k]\right)^{\mathrm{T}} \boldsymbol{\Gamma} \left(\bar{\boldsymbol{\Delta u}}_S^R[k]\right) + \boldsymbol{\gamma}^{\mathrm{T}} \bar{\boldsymbol{\Delta u}}_S^R[k] \\
&= \arg\min_{\bar{\boldsymbol{\Delta u}}_S} \; \frac{1}{2}\|\bar{\boldsymbol{\Delta u}}_S^R[k]\|_{\boldsymbol{\Gamma}}^2 + \boldsymbol{\gamma}^{\mathrm{T}} \bar{\boldsymbol{\Delta u}}_S^R[k] \\
\text{s.t.} \quad & \underbrace{\begin{bmatrix} \boldsymbol{M}_u \\ \boldsymbol{M}_i \end{bmatrix}}_{\boldsymbol{M}} \bar{\boldsymbol{\Delta u}}_S^R[k] \leq \underbrace{\begin{bmatrix} \boldsymbol{\mu}_u \\ \boldsymbol{\mu}_i \end{bmatrix}}_{\boldsymbol{\mu}}
\end{aligned}
\tag{3.6}
$$

with the Hessian matrix of the cost function being positive definite $\nabla^2_{\bar{\boldsymbol{\Delta u}}[k]} J = \boldsymbol{\Gamma} \succ 0$ to ensure convexity.

3.2.1 Cost Function

For the cost function J, a simple LQR-style base cost is employed. With $\boldsymbol{Q} \in \mathbb{R}^{N_P q \times N_P q}$ as the diagonal weighting matrix for the tracking error and $\boldsymbol{R} \in \mathbb{R}^{N_P m \times N_P m}$ the diagonal weighting matrix for the stator voltage increment. For the application of electric drives, reference tracking is significantly more important then the magnitude of the stator voltage.

$$
J = \left(\, \bar{\boldsymbol{i}}[k+1] - \bar{\boldsymbol{i}}_{ref}[k+1] \,\right)^{\mathrm{T}} \boldsymbol{Q} \left(\, \bar{\boldsymbol{i}}[k+1] - \bar{\boldsymbol{i}}_{ref}[k+1] \,\right) + \left(\bar{\boldsymbol{\Delta u}}[k]\right)^{\mathrm{T}} \; \boldsymbol{R} \, \bar{\boldsymbol{\Delta u}}[k]
$$

Or with the more compact $\mathcal{L}_2$-norm notation $\|\boldsymbol{x}\|_{\boldsymbol{I}}^2 = \boldsymbol{x}^{\mathrm{T}}\boldsymbol{I}\boldsymbol{x}$:

$$
J = \|\bar{\boldsymbol{i}} - \bar{\boldsymbol{i}}_{ref}\|_Q^2 + \|\bar{\boldsymbol{\Delta u}}\|_{\boldsymbol{R}}^2 \tag{3.7}
$$

To express the cost J in terms of $\bar{\boldsymbol{\Delta u}}[k]$, the prediction for the current as output signal is expressed by the prediction based on the system $\Sigma : [\boldsymbol{A}(\omega[k]), \boldsymbol{B}, \boldsymbol{E}(\omega[k])]$ according to Equation (3.5).

$$\bar{\boldsymbol{i}}[k+1] = \bar{\boldsymbol{y}}[k+1] = \bar{\boldsymbol{g}}[k] + \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k]$$

A prediction of the reference current is given as input $\bar{\boldsymbol{i}}_{ref}[k+1] = \bar{\boldsymbol{r}}[k+1]$.

$$\begin{aligned}
\bar{\boldsymbol{i}}[k+1] - \bar{\boldsymbol{i}}_{ref}[k+1] &= \bar{\boldsymbol{y}}[k+1] - \bar{\boldsymbol{r}}[k+1] \\
&= \underbrace{\boldsymbol{P}_x \boldsymbol{x} + \boldsymbol{P}_u \boldsymbol{u}[-1] + \boldsymbol{P}_E \boldsymbol{E}(\omega[k])}_{\bar{\boldsymbol{g}}[k]} - \bar{\boldsymbol{r}}[k+1] + \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k] \\
&= \tilde{\bar{\boldsymbol{e}}}[k+1] + \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k]
\end{aligned}$$

Here, $\tilde{\bar{\boldsymbol{e}}}[k+1] = \bar{\boldsymbol{g}}[k] - \bar{\boldsymbol{r}}[k+1]$ is defined as the prediction of the quasi-tracking error. This is a summation of the prediction terms independent of $\bar{\boldsymbol{\Delta u}}[k]$. The separation is done in order to use $\bar{\boldsymbol{\Delta u}}[k]$ as the vector of optimization variables. Inserting this into (3.7), leads to:

$$J = \left(\tilde{\bar{\boldsymbol{e}}}[k+1] + \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k]\right)^T \boldsymbol{Q} \left(\tilde{\bar{\boldsymbol{e}}}[k+1] + \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k]\right) + \left(\bar{\boldsymbol{\Delta u}}[k]\right)^{\mathrm{T}} \boldsymbol{R}\, \bar{\boldsymbol{\Delta u}}[k]$$

For readability the indices k for the quasi-error prediction are not written down. Using the symmetric property of the diagonal weighting matrices $\boldsymbol{Q}, \boldsymbol{R}$, the cost function is expanded to:

$$\begin{aligned}
J = {} & \tilde{\bar{\boldsymbol{e}}}^T \boldsymbol{Q} \tilde{\bar{\boldsymbol{e}}} + 2\left(\bar{\boldsymbol{\Delta u}}[k]\right)^T \boldsymbol{P}_{\bar{\Delta u}}^T \boldsymbol{Q} \tilde{\bar{\boldsymbol{e}}} \\
& + \left(\bar{\boldsymbol{\Delta u}}[k]\right)^T \boldsymbol{P}_{\bar{\Delta u}}^T \boldsymbol{Q} \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k] + \left(\bar{\boldsymbol{\Delta u}}[k]\right)^T \boldsymbol{R} \bar{\boldsymbol{\Delta u}}[k] \\
= {} & \left(\bar{\boldsymbol{\Delta u}}[k]\right)^T \left(\boldsymbol{P}_{\bar{\Delta u}}^T \boldsymbol{Q} \boldsymbol{P}_{\bar{\Delta u}} + \boldsymbol{R}\right) \bar{\boldsymbol{\Delta u}}[k] + 2\tilde{\bar{\boldsymbol{e}}}^T \boldsymbol{Q} \boldsymbol{P}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k] \\
& + \tilde{\bar{\boldsymbol{e}}}^T \boldsymbol{Q} \tilde{\bar{\boldsymbol{e}}}.
\end{aligned}$$

Due to $\tilde{\bar{\boldsymbol{e}}}^T \boldsymbol{Q} \tilde{\bar{\boldsymbol{e}}}$ being constant in relation to the optimization vector, it can be ignored. Using the aforementioned compact notation and coefficient comparison for the matrices of the Quadratic Program (3.6) can be conducted.

$$\begin{aligned} J = \quad & \|\bar{\Delta u}[k]\|^2_{(P_{\bar{\Delta u}})^{\mathrm{T}} Q P_{\bar{\Delta u}} + R} && + \quad 2\left(\tilde{e}[k+1]\right)^{\mathrm{T}} Q P_{\bar{\Delta u}} \bar{\Delta u}[k] \\ \overset{!}{=} \quad & \frac{1}{2}\|\bar{\Delta u}[k]\|^2_{\Gamma} && + \quad \gamma^{\mathrm{T}} \bar{\Delta u}[k] \end{aligned}$$

Keeping the factor 1/2 for the quadratic term in mind, the matrices Γ, γ are defined as

$$\Gamma = 2\left((P_{\bar{\Delta u}})^{\mathrm{T}} Q P_{\bar{\Delta u}} + R\right) \qquad \gamma^{\mathrm{T}} = 2\left(\tilde{e}[k+1]\right)^{\mathrm{T}} Q P_{\bar{\Delta u}}$$

To ensure convexity, Q, R must be chosen such that $\Gamma \succ 0$.

3.2.2 Voltage Constraints

As mentioned in Section **??**, the stator voltage vector u_S^S is subject to the constraints of the voltage hexagon spanned by the 6 active vectors. This is depicted in Figure 3.1.

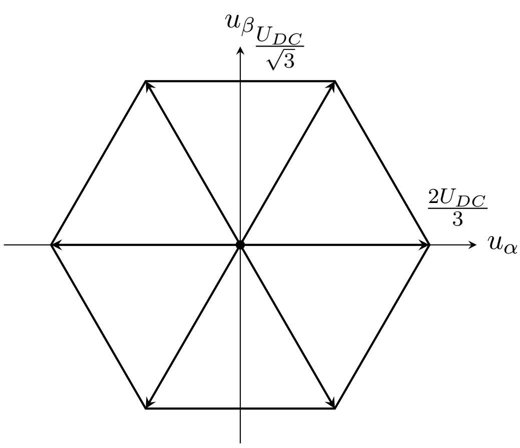


Figure 3.1: Voltage Hexagon in SRF

These constraints can be formulated as a set of linear inequality constraints. For example, the upper flat border of the hexagon is described by

$$u_\beta \leq \frac{U_{DC}}{\sqrt{3}} \tag{3.8}$$

and the upper right border, with a slope of $-\pi/3$ by

$$u_\beta \leq -\sqrt{3}u_\alpha + \frac{2\,U_{DC}}{\sqrt{3}} \tag{3.9}$$

Both of the inequalities (3.8) and (3.9) describe convex sets. Continuing this construction of the voltage hexagon for all six borders, leads to a sixfold convex intersection of half-planes as in [8] [9], which describes the voltage hexagon. Combining those six relations leads to the set of Linear Inequalities in SRF (3.10).

$$\underbrace{\begin{bmatrix} \sqrt{3} & 1 \\ 0 & 1 \\ -\sqrt{3} & 1 \\ -\sqrt{3} & -1 \\ 0 & -1 \\ \sqrt{3} & -1 \end{bmatrix}}_{\boldsymbol{N}_u^S} \begin{bmatrix} u_\alpha \\ u_\beta \end{bmatrix} \leq \underbrace{\frac{U_{DC}}{\sqrt{3}} \begin{bmatrix} 2 \\ 1 \\ 2 \\ 2 \\ 1 \\ 2 \end{bmatrix}}_{\boldsymbol{\nu}_u^S} \tag{3.10}$$

Using Equation (3.10), the voltage constraints in the stator reference frame can be compactly written down as:

$$\boldsymbol{N}_u^S \boldsymbol{u}_S^S[k] \leq \boldsymbol{\nu}_{\boldsymbol{u}}^{\boldsymbol{S}}$$

However, the optimization problem shall be defined with respect to the the rotor reference frame in terms of the incremental stator voltage predictions. To do so, first, the rotation from the RRF is applied to $\boldsymbol{u}_S^R[k] = \boldsymbol{\Delta u}_S^R[k] + \boldsymbol{u}_S^R[k-1]$ for one voltage vector.

$$\boldsymbol{N}_u^S \boldsymbol{T}_{dq\to\alpha\beta}(\varphi[k]) \left(\boldsymbol{\Delta u}_S^R[k] + \boldsymbol{u}_S^R[k-1]\right) \leq \boldsymbol{\nu}_{\boldsymbol{u}}^{\boldsymbol{S}}$$

The next step is to extend that formulation to the incremental prediction vector $\boldsymbol{\Delta u}_S^R[k] \to \bar{\boldsymbol{\Delta u}}_S^R[k]$. To calculate $\boldsymbol{u}_S^R[k+1] \dots \boldsymbol{u}_S^R[k+N_P-1]$ from $\boldsymbol{u}_S^R[k-1]$ and $\boldsymbol{\Delta u}_S^R[k+l]$, all the $\boldsymbol{\Delta u}_S^R[k+l]$ with $l \leq N_P - 1$ must be accumulated. Thus, the prediction vector in RRF is multiplied by the lower block-triangular matrix $\boldsymbol{P}_T$ with $\mathbf{I}_2$ as block elements.

$$\bar{\boldsymbol{u}}[k] = \boldsymbol{P}_T \bar{\boldsymbol{\Delta u}}[k] + (\mathbf{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{u}[k-1]$$

With $(\mathbf{1}_{N_P} \otimes \boldsymbol{I}_2)$ being the Kronecker product, describing N_P stacked identity matrices. Putting everything together leads to:

$$\bar{\boldsymbol{N}}_u^S \bar{\boldsymbol{T}}_{dq\to\alpha\beta}(\bar{\varphi}[k]) \left(\boldsymbol{P}_T \bar{\boldsymbol{\Delta u}}_S^R[k] + (\mathbf{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{u}_S^R[k-1]\right) \leq \bar{\boldsymbol{\nu}}_u^S \tag{3.11}$$

Or in more detail:

$$\underbrace{\begin{bmatrix} \boldsymbol{N}_u^S & \boldsymbol{0}^{6\times 2} & \cdots & \boldsymbol{0}^{6\times 2} \\ \boldsymbol{0}^{6\times 2} & \boldsymbol{N}_u^S & \cdots & \boldsymbol{0}^{6\times 2} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0}^{6\times 2} & \boldsymbol{0}^{6\times 2} & \cdots & \boldsymbol{N}_u^S \end{bmatrix}}_{\bar{\boldsymbol{N}}_u^S} \underbrace{\begin{bmatrix} \boldsymbol{T}_{dq\to\alpha\beta}(\varphi[k]) & \boldsymbol{0}_2 & \cdots & \boldsymbol{0}_2 \\ \boldsymbol{0}_2 & \boldsymbol{T}_{dq\alpha\beta}\left(\varphi[k+1]\right) & \cdots & \boldsymbol{0}_2 \\ \vdots & \vdots & \ddots & \boldsymbol{0}_2 \\ \boldsymbol{0}_2 & \boldsymbol{0}_2 & \cdots & \boldsymbol{T}_{dq\alpha\beta}\left(\varphi[k+N_P-1]\right) \end{bmatrix}}_{\bar{\boldsymbol{T}}_{dq\to\alpha\beta}(\bar{\varphi}[k])} \cdot$$

$$\left(\underbrace{\begin{bmatrix} \boldsymbol{I}_2 & \boldsymbol{0}_2 & \boldsymbol{0}_2 & \cdots & \boldsymbol{0}_2 \\ \boldsymbol{I}_2 & \boldsymbol{I}_2 & \boldsymbol{0}_2 & \cdots & \boldsymbol{0}_2 \\ \vdots & \vdots & \ddots & & \boldsymbol{0}_2 \\ \boldsymbol{I}_2 & \boldsymbol{I}_2 & \boldsymbol{I}_2 & \cdots & \boldsymbol{I}_2 \end{bmatrix}}_{\boldsymbol{P}_T} \underbrace{\begin{bmatrix} \boldsymbol{\Delta u}[k] \\ \boldsymbol{\Delta u}[k+1] \\ \vdots \\ \boldsymbol{\Delta u}[k+N_P-1] \end{bmatrix}}_{\bar{\boldsymbol{\Delta u}}_S^R[k]} + \begin{bmatrix} \boldsymbol{I}_2 \\ \boldsymbol{I}_2 \\ \vdots \\ \boldsymbol{I}_2 \end{bmatrix} \boldsymbol{u}_S^R[k-1] \right) \leq \begin{bmatrix} \boldsymbol{\nu}_u^S \\ \boldsymbol{\nu}_u^S \\ \vdots \\ \boldsymbol{\nu}_u^S \end{bmatrix}$$

As a final step, to arrive at a set of linear inequalities in terms of $\bar{\boldsymbol{\Delta u}}[k]$, Equation (3.11) must be rearranged to:

$$\bar{\boldsymbol{N}}_u^S \bar{\boldsymbol{T}}_{dq\to\alpha\beta}(\bar{\varphi}[k]) \boldsymbol{P}_T \boldsymbol{\Delta u}_S^R[k] \leq \bar{\boldsymbol{\nu}}_u^S - \bar{\boldsymbol{N}}_u^S \bar{\boldsymbol{T}}_{dq\to\alpha\beta}(\bar{\varphi}[k])(\boldsymbol{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{u}_S^R[k-1]$$

$\boldsymbol{T}_{dq\to\alpha\beta}(\varphi[k])$ is part of the SO_2 group (Special-Orthogonal in the plane $\mathbb{R}^2$) [10] [11]. Thus, using the assumption $\omega[k] = \omega[k+l] \ : \ l \in [1, N_P]$ for $\varphi[k+l] = \varphi[k] + l \cdot \omega[k] T_S$ the transformation matrices for future steps can be approximated by the following product:

$$\boldsymbol{T}_{dq\alpha\beta}\left(\varphi[k+l]\right) = \left(\boldsymbol{T}_{dq\to\alpha\beta}\left(\omega[k]T_S\right)\right)^l \boldsymbol{T}_{dq\to\alpha\beta}(\varphi[k])$$

To further reduce the computational burden, the assumption $\omega[k]T_S \leq \omega_{max} T_{S,max} \leq 0.2$ holds true $\forall k$ of electric drives considered within the scope of this thesis. This allows for small angle approximation.

$$\boldsymbol{T}_{dq\to\alpha\beta}\left(\omega[k]T_S\right) \approx \begin{bmatrix} 1 & -\omega[k]T_S \\ \omega[k]T_S & 1 \end{bmatrix} := \tilde{\boldsymbol{T}}_{dq\alpha\beta}(\omega[k]T_S)$$

With $\bar{\tilde{\boldsymbol{T}}}_{dq\alpha\beta}(\bar{\varphi}[k])$ being an block diagonal matrix of elements $\tilde{\boldsymbol{T}}_{dq\alpha\beta}(\omega[k]T_S)$, the following computational-effective implementation can be written down.

$$\underbrace{\bar{\boldsymbol{N}}_u^S \bar{\tilde{\boldsymbol{T}}}_{dq\alpha\beta}(\bar{\varphi}[k]) \boldsymbol{P}_T}_{\boldsymbol{M}_u} \boldsymbol{\Delta u}_S^R[k] \leq \underbrace{\bar{\boldsymbol{\nu}}_u^S - \bar{\boldsymbol{N}}_u^S \bar{\tilde{\boldsymbol{T}}}_{dq\alpha\beta}(\bar{\varphi}[k])(\boldsymbol{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{u}_S^R[k-1]}_{\boldsymbol{\mu}_u} \tag{3.12}$$

3.2.3 Current Constraints

To ensure safe operation, the current space vector must not exceed its limit I_{max}. Due to the rotational symmetry of this constraint, the formulation is equivalent for SRF and RRF.

$$\|\boldsymbol{i}_S^S\|_2 \leq I_{max} \iff \|\boldsymbol{i}_S^R\|_2 \leq I_{max} \tag{3.13}$$

In order to allow for short-term violation of this constraint, comparable to I^2t safety protection schemes, soft constraints are introduced in Section 3.5.4.
The Constraint 3.13 describes a disk on the complex plane. In a lot MPC literature this is approximated by a polygon, to allow for a constraint description as a finite set of linear inequality constraints. However, this usually leads to a high amount of inequalities [9]. One approach to reduce that, it so introduce parameter-varying constraints, as in [4].

Parameter-Invariant Constraints

Constraint (3.13) was approximated by an octagon. The octagon is oriented in a way, such that the $\pm i_q$ under approximated constraints are exactly equal I_{max} for $i_d = 0$. This is depicted in Figure 3.2.

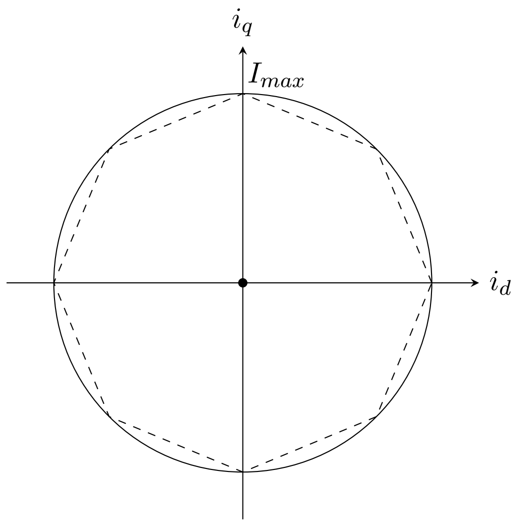


Figure 3.2: Current constraints approximation in RRF

Analog to the procedure described in Section 3.2.2 for the voltage hexagon, an octagon is constructed. For the full analytic derivation of the set of inequalities for the octagon, refer to Section 5.2 of [9].

$$\underbrace{\begin{bmatrix} 1 & \frac{1-c}{c} \\ \frac{1-c}{c} & 1 \\ -\frac{1-c}{c} & 1 \\ -1 & \frac{1-c}{c} \\ -1 & -\frac{1-c}{c} \\ -\frac{1-c}{c} & -1 \\ \frac{1-c}{c} & -1 \\ 1 & -\frac{1-c}{c} \end{bmatrix}}_{\boldsymbol{N}_i^R} \begin{bmatrix} i_d \\ i_q \end{bmatrix} \leq \underbrace{\begin{bmatrix} I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \\ I_{max} \end{bmatrix}}_{\boldsymbol{\nu}_i^R}$$

With the auxiliary variable $c = \cos \pi/4$. The constraints regarding the stator current can be written down as:

$$\boldsymbol{N}_i^R \boldsymbol{i}_S^R[k+1] \leq \boldsymbol{\nu}_i^R$$

To arrive at a set of inequalities in terms of $\boldsymbol{\Delta u}$ for the Quadratic Program (3.6), $\boldsymbol{i}_S^R[k+1]$ is expressed by $\Sigma : [\boldsymbol{A}(\omega[k]), \boldsymbol{B}, \boldsymbol{E}(\omega[k])]$ with $\boldsymbol{u}_S^R[k] = \boldsymbol{u}_S^R[k-1] + \boldsymbol{\Delta u}_S^R[k]$

$$\boldsymbol{N}_i^R \left(\boldsymbol{A}(\omega[k]) \boldsymbol{i}_S^R[k] + \boldsymbol{B} \left(\boldsymbol{u}_S^R[k-1] + \boldsymbol{\Delta u}_S^R[k] \right) + \boldsymbol{E}(\omega[k]) \right) \leq \boldsymbol{\nu_i^R}$$

Using the Prediction Equation (3.5), the constraints for the prediction vector $\bar{\boldsymbol{\Delta u}}[k]$ can be derived

$$\bar{\boldsymbol{N}}_i^R \left(\bar{\boldsymbol{g}}[k] + \boldsymbol{P}_{\bar{\Delta u}}(\omega[k]) \bar{\boldsymbol{\Delta u}}_S^R[k] \right) \leq \bar{\boldsymbol{\nu}}_i^R$$

or in more detail:

$$\underbrace{\begin{bmatrix} \boldsymbol{N}_i^R & \boldsymbol{0}^{8\times 2} & \cdots & \boldsymbol{0}^{8\times 2} \\ \boldsymbol{0}^{8\times 2} & \boldsymbol{N}_i^R & \cdots & \boldsymbol{0}^{8\times 2} \\ \vdots & \vdots & \ddots & \vdots \\ \boldsymbol{0}^{8\times 2} & \boldsymbol{0}^{8\times 2} & \cdots & \boldsymbol{N}_i^R \end{bmatrix}}_{\bar{\boldsymbol{N}}_i^R} \left(\underbrace{\begin{bmatrix} \boldsymbol{g}[k] \\ \boldsymbol{g}[k+1] \\ \vdots \\ \boldsymbol{g}[k+N_P-1] \end{bmatrix}}_{\bar{\boldsymbol{g}}[k]} + \boldsymbol{P}_{\bar{\Delta u}}(\omega[k]) \underbrace{\begin{bmatrix} \Delta \boldsymbol{u}[k] \\ \Delta \boldsymbol{u}[k+1] \\ \vdots \\ \Delta \boldsymbol{u}[k+N_P-1] \end{bmatrix}}_{\bar{\boldsymbol{\Delta u}}_S^R[k]} \right) \leq \underbrace{\begin{bmatrix} \boldsymbol{\nu}_i^R \\ \boldsymbol{\nu}_i^R \\ \vdots \\ \boldsymbol{\nu}_i^R \end{bmatrix}}_{\bar{\boldsymbol{\nu}}_i^R}$$

This equation is the rearranged in terms of $\boldsymbol{M}_i \bar{\boldsymbol{\Delta u}}_S^R[k] \leq \boldsymbol{\mu}_i$:

$$\underbrace{\bar{\boldsymbol{N}}_i^R \boldsymbol{P}_{\bar{\Delta u}}}_{\boldsymbol{M}_i} \bar{\boldsymbol{\Delta u}}_S^R[k] \leq \underbrace{\bar{\boldsymbol{\nu}}_i^R - \bar{\boldsymbol{N}}_i^R \bar{\boldsymbol{g}}[k]}_{\boldsymbol{\mu}_i}$$

Reduced Order Time-Variant Constraints

Another idea is proposed in the context fo the time-optimal model predictive flux controller from a group of researches at the University of Paderborn [4]. Here, the circular current constraint are approximated by a tangent normal to the time-variant current space vector $\boldsymbol{i}_S^R$ as seen in Figure 3.3. The radius of that constraint is dynamically set at an higher level by an operating point controller, such that for short-term transient events are allowed to violate the steady-state maximum current $I_{max,dyn}[k] \geq I_{max}$. This thesis explores an approach for dynamic current constraints based on a slackness variable (See Section 3.5.4).

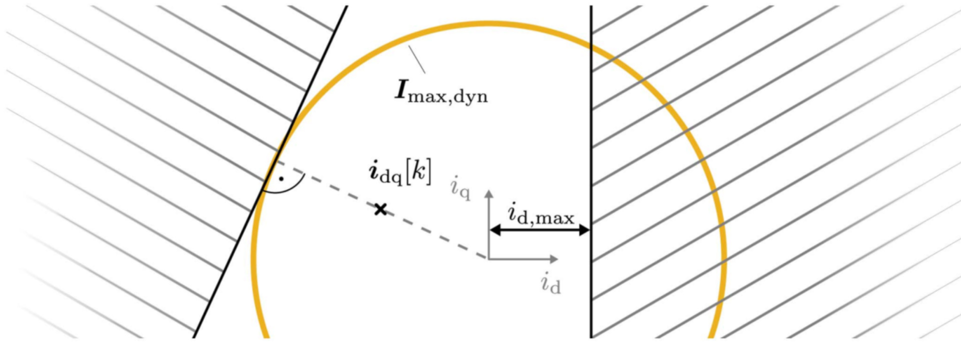


Figure 3.3: Linear time-variant current constraints. Image source: [4]

A seen in Figure 3.3, an additional time-variant halfspace constraint is introduced, which aims to mitigates nonmonotic torque due to positive reference i_d currents. [4]

3.2.4 Hildreth's Quadratic Programming Procedure

The Quadratic Program (3.6) is solved using Hildreth's Quadratic Programming Procedure, as recommended in [12], due to its benefits for real-time optimization problems. The iterative solver employs the basis vectors as search directions, in contrast to the Newton approach, which determines a search direction based on the Hessian matrix of the cost function, $\nabla^2 J = \boldsymbol{\Gamma}$. Therefore Hildreth's procedure performs an element-by-element search, where only one Lagrange multiplier is updated at each step.
Some of the key advantages of Hildreth's procedure are:

Solution improves monotonically : If a feasable solution exists, successive iterations are guaranteed converge towards the optimum.

Fixed computational effort : The number of iterations can be limited due the monotonic improvement.

Element by element search : No matrix inversion is required. This reduces the computational burden.

For a detailed description of this procedure in MPC applications refer to the MPC Book by L. Wang [12].

3.3 MPC Formulations

For field-oriented current control applications, three different MPC formulations are compared. The first one, referred to here as the Basic MPC, is based on [7] [3] and is often used in the literature due to its straightforward implementation.
The second approach utilizes an augmented incremental state-space system in order to embed an integrator directly into the controller structure, as proposed in [12].
The third can be thought of as a hybrid approach combining the previous two formulations, in order to obtain an embedded integrator while retaining a similar implementation to the Basic MPC. As will be shown, this is beneficial for extensions, such as those described in Section 3.5.
It is worth mentioning that, for LTI systems, so-called offset-free MPC can be used to guarantee the rejection of constant disturbances. In the classical approach this is achieved by estimating the disturbances as an additional state with a constant dynamic $\hat{d}[k+1] = \hat{d}[k]$. For reference, see chapter 5 of [3] or the summary paper [13] , which builds on the original work presented in [14]. However, since a nonlinear prediction model was chosen for this application, some of the key assumptions made in those publications cannot be satisfied. This problem was recognized and the approach presented in [6] mitigates this by estimating multiple disturbances for the application of FOC for PMSM.

3.3.1 Basic MPC

The prediction according to Equation (3.5) uses absolute states $\boldsymbol{x}[k]$ and differential actuating signal $\boldsymbol{\Delta u}[k] = \boldsymbol{u}[k] - \boldsymbol{u}[k-1]$. This enables the utilization of a commonly used MPC structure, as described in Section 2.3 of [7], which is depicted in Figure 3.4.
The MPC is provided with a reference prediction

$$\bar{\boldsymbol{r}}[k+1] = \left[\boldsymbol{r}^{\mathrm{T}}[k+1]\ \boldsymbol{r}^{\mathrm{T}}[k+2]\ \dots \boldsymbol{r}^{\mathrm{T}}[k+N_P]\right]^{\mathrm{T}}$$

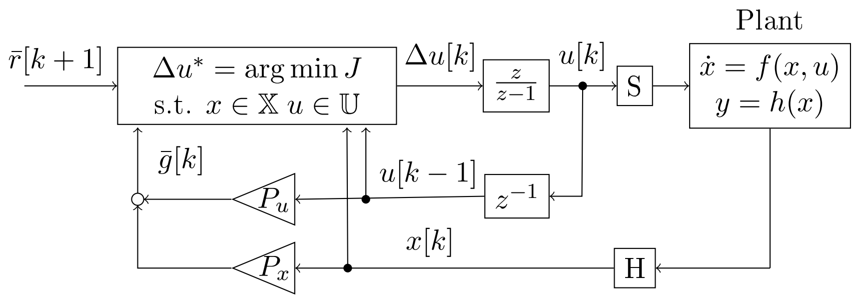


Figure 3.4: Basis MPC structure used in [7] with the notation used in this thesis

. Using current states and the previous input, the auxiliary prediction vector $\bar{\boldsymbol{g}}[k]$ can be computed according to Equation 3.5. Using $\tilde{\bar{\boldsymbol{e}}}[k+1] = \bar{\boldsymbol{g}}[k] - \bar{\boldsymbol{r}}[k+1]$ to calculate the linear term $\boldsymbol{\gamma}$ of the quadratic program, the cost function $J = \|\boldsymbol{\Delta u}\|_{\boldsymbol{\Gamma}}^2 + \boldsymbol{\gamma}^{\boldsymbol{T}}\bar{\boldsymbol{\Delta u}}[k]$ is fully determined. To determine the constraints $\boldsymbol{M}\bar{\boldsymbol{\Delta u}}[k] \leq \boldsymbol{\mu}$, measured state vector and previous input are required. The output of the Quadratic Program (3.6) is the incremental input $\bar{\boldsymbol{\Delta u}}[k]$, which is accumulated then to $\boldsymbol{u}[k]$. The interface to the plant is modeled as Sample and Hold elements.
Section 3.4 provides an analysis on why this approach does not inherently guarantee offset-free tracking, but can be extended by using the approach discussed in Section 3.5.2.

3.3.2 MPC with Embedded Integrator

A different approach is used by L. Wang [12]. Here, the state space system is augmented, such that the states and actuating signals are incremental. As it will be shown in Section 3.4, this ensures the MPC is of integrating behavior for locally unconstrained operation. In this thesis a brief overview of the derivation of the augmented state space model is given. For more details, refer to Section 1.2 of the MPC Book by Wang [12].
For discrete-time, linear time-invariant systems, without feedthrough term $\Sigma : [\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}]$, the incremental states and inputs are defined as:

$$\boldsymbol{\Delta x}[k] = \boldsymbol{x}[k] - \boldsymbol{x}[k-1] \qquad\qquad \boldsymbol{\Delta u}[k] = \boldsymbol{u}[k] - \boldsymbol{u}[k-1]$$

In order to predict the absolute state or output $\boldsymbol{y}$, an additional state is introduced that accumulates the incremental quantities. This leads to the following augmented state-space model:

$$\begin{aligned}
\begin{bmatrix} \boldsymbol{\Delta x}[k+1] \\ \boldsymbol{y}[k+1] \end{bmatrix} &= \begin{bmatrix} \boldsymbol{A} & \boldsymbol{0} \\ \boldsymbol{CA} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Delta x}[k] \\ \boldsymbol{y}[k] \end{bmatrix} + \begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{CB} \end{bmatrix} \boldsymbol{\Delta u}[k] \\
\boldsymbol{y}[k] &= \begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \end{bmatrix} \begin{bmatrix} \boldsymbol{\Delta x}[k] \\ \boldsymbol{y}[k] \end{bmatrix}
\end{aligned} \tag{3.14}$$

Applying this to the description of the affine, parameter-varying discrete time model used for the PMSM (3.3), one has to take especially care for the the ω-varying terms:

$$\begin{aligned}
\Sigma_{k+1} : [\boldsymbol{A}(\omega[k]), \boldsymbol{B}, \boldsymbol{E}(\omega[k])] - \Sigma_k : [\boldsymbol{A}(\omega[k-1]), \boldsymbol{B}, \boldsymbol{E}(\omega[k-1])] &= \\
\boldsymbol{x}[k+1] - \boldsymbol{x}[k] &= (\boldsymbol{A}(\omega[k])\,(\boldsymbol{x}[k-1] + \boldsymbol{\Delta x}[k]) + \boldsymbol{Bu}[k] + \boldsymbol{E}(\omega[k])) \\
&\quad - (\boldsymbol{A}(\omega[k-1])\boldsymbol{x}[k-1] + \boldsymbol{Bu}[k-1] + \boldsymbol{E}(\omega[k-1])) \\
\boldsymbol{\Delta x}[k+1] &= \boldsymbol{A}(\omega[k])\boldsymbol{\Delta x}[k] + \boldsymbol{B} \cdot \boldsymbol{\Delta u}[k] + \underbrace{\boldsymbol{\Delta A}(\Delta\omega[k])\boldsymbol{x}[k-1] + \boldsymbol{\Delta E}(\Delta\omega[k])}_{\boldsymbol{F}[k]}
\end{aligned}$$

with

$$\boldsymbol{\Delta A}(\Delta\omega[k]) = \begin{bmatrix} \boldsymbol{0} & T_s \dfrac{L_q}{L_d}\Delta\omega[k] \\ -T_s \dfrac{L_d}{L_q}\Delta\omega[k] & \boldsymbol{0} \end{bmatrix} \quad \boldsymbol{\Delta E}(\Delta\omega[k]) = \begin{bmatrix} 0 \\ -\frac{T_s \psi_{PM}}{L_q}\Delta\omega[k] \end{bmatrix}.$$

Applying the incremental state accumulation as in Equation (3.14), the augmented state space system for the PMSM can be derived:

$$\begin{aligned}
\underbrace{\begin{bmatrix} \boldsymbol{\Delta x}[k+1] \\ \boldsymbol{y}[k+1] \end{bmatrix}}_{\boldsymbol{\xi}[k+1]} &= \underbrace{\begin{bmatrix} \boldsymbol{A}(\omega[k]) & \boldsymbol{0} \\ \boldsymbol{CA}(\omega[k]) & \boldsymbol{I} \end{bmatrix}}_{\boldsymbol{A}_\Delta(\omega[k])} \underbrace{\begin{bmatrix} \boldsymbol{\Delta x}[k] \\ \boldsymbol{y}[k] \end{bmatrix}}_{\boldsymbol{\xi}[k]} + \underbrace{\begin{bmatrix} \boldsymbol{B} \\ \boldsymbol{CB} \end{bmatrix}}_{\boldsymbol{B}_\Delta} \boldsymbol{\Delta u}[k] + \underbrace{\begin{bmatrix} \boldsymbol{F}[k] \\ \boldsymbol{CF}[k] \end{bmatrix}}_{\boldsymbol{E}_\Delta[k]} \\
\boldsymbol{y}[k] &= \underbrace{\begin{bmatrix} \boldsymbol{0} & \boldsymbol{I} \end{bmatrix}}_{\boldsymbol{C}_\Delta} \begin{bmatrix} \boldsymbol{\Delta x}[k] \\ \boldsymbol{y}[k] \end{bmatrix}
\end{aligned} \tag{3.15}$$

Apart from the ambiguity regarding prediction assumptions for $\Delta\omega$, implementing constraints or extensions (Section 3.5) with incremental quantities for states, $\boldsymbol{A}(\omega[k])$ and $\boldsymbol{E}(\omega[k])$ proved to introduce additional complexity.

For the implementation of this augmented system, one has to obtain the incremental state change $\boldsymbol{\Delta x}[k]$. This is accomplished by the filter $1 - z^{-1}$. To calculate the constraints and tracking error, $u[k-1]$ and $x[k]$ are needed. This results into the controller structure shown in Figure 3.5.

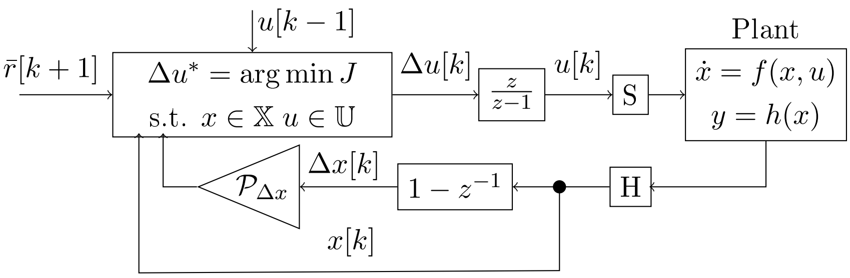


Figure 3.5: MPC structure used in [12] with the notation used in this thesis

Where $\mathcal{P}_{\Delta x}$ is the prediction matrices for the incremental state feedback of the augmented system Σ_Δ, similar to the structure of $\boldsymbol{P}_x$ as given by Equation (3.5). The advantage of this approach, which introduces integrating behavior, is accompanied by the additional complexity due incremental states for prediction.

3.3.3 Hybrid Approach

Since the basic MPC formulation does not inherently guarantee offset-free tracking under constant disturbances, such as model faults and the embedded integrator according to Wang hinders extensions and constraint formulations due to its incremental states, a way searched to combine the benefits of those two approaches.
To obtain an MPC using absolute states $\boldsymbol{x}[k] = [i_d \; i_q]^{\mathrm{T}}$ and still having no direct feedback for $\boldsymbol{u}[k-1]$ for the locally unconstrained controller, like in the Embedded Integrator structure of, a hybrid structure of these two approaches was designed. The result is depicted in Figure 3.6.
The idea behind this structure is to use the state description to estimate what input $\tilde{\boldsymbol{u}}[k-1]$ was applied to the plant such that it leads to the measured states $x[k]$. By doing so, input disturbances $\boldsymbol{d}_1[k-1] = \tilde{\boldsymbol{u}}[k-1] - \boldsymbol{u}[k-1]$ are considered.

$$\boldsymbol{x}[k] = \boldsymbol{A}\boldsymbol{x}[k-1] + \boldsymbol{B}\underbrace{(\boldsymbol{d}_1[k-1] + \boldsymbol{u}[k-1])}_{\tilde{\boldsymbol{u}}[k-1]}$$

$$\tilde{\boldsymbol{u}}[k-1] = \boldsymbol{B}^{-1}\left(\boldsymbol{x}[k] - \boldsymbol{A}\boldsymbol{x}[k-1]\right).$$

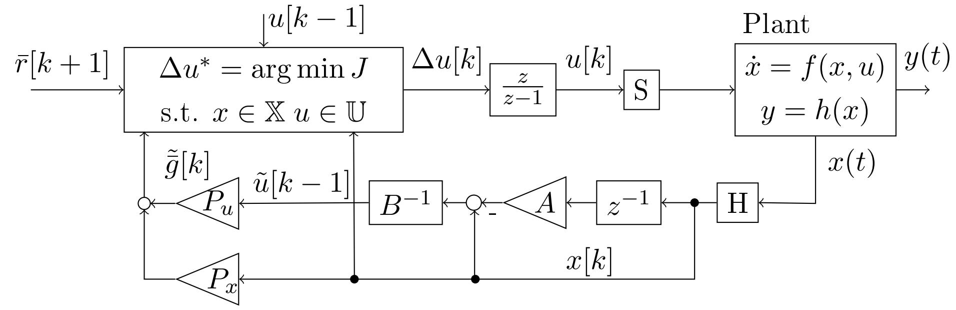


Figure 3.6: Hybrid MPC structure for LTI systems

Generally speaking, the inverse $\boldsymbol{B}^{-1}$ is not guaranteed to exist. However, in the case of the PMSM model used (3.3), $\boldsymbol{B}$ is a diagonal matrix with positive elements. With L_d, L_q being in the range of $[0.1, 10]$ mH, $\boldsymbol{B}$ is well conditioned.
For the affine, parameter-varying prediction model $\boldsymbol{E}(\omega[k])$ has to be considered as well:

$$\tilde{\boldsymbol{u}}[k-1] = \boldsymbol{B}^{-1} \left(\boldsymbol{x}[k] - \boldsymbol{A}(\omega[k-1])\boldsymbol{x}[k-1] - \boldsymbol{E}(\omega[k-1]) \right).$$

The dynamic behavior of this approach will be analyzed and compared to the other two formulations in the Section 3.4.

3.4 Analysis of the Controller Structure

The main objective of this Section is to analyze the effect of constant disturbances such as model faults on the reference tracking of the controller. Since for the scope of this thesis, the stator current relation of the PMSM according to Equation (2.3), will be the only plant considered it is deemed sufficient to only analyze the three different controller structures.

3.4.1 Closed-Loop Structure with inactive Constraints

For the unconstrained case, the Quadratic Program can be solved offline by finding the stationary points of J with respect to $\bar{\boldsymbol{\Delta u}}[k]$:

$$\begin{aligned} \nabla_{\bar{\boldsymbol{\Delta u}}[k]} J &= \boldsymbol{\Gamma} \bar{\boldsymbol{\Delta u}}[k] + \boldsymbol{\gamma} = 0 \\ \boldsymbol{\Delta u}^*[k] &= -\boldsymbol{\Gamma}^{-1} \boldsymbol{\gamma} \end{aligned}$$

With the Hessian matrix $\nabla^2_{\bar{\boldsymbol{\Delta u}}[k]} J = \boldsymbol{\Gamma} \succ 0$ ensuring the stationary point to be a minimizer and $\boldsymbol{\Gamma} \succ 0$ is ensured by the choice of $\boldsymbol{Q}, \boldsymbol{R}$.

In model predictive control only the first signal of the computed optimal input prediction is applied, thus the locally unconstrained linear control law $\bar{\boldsymbol{\Delta u}}[k] = \boldsymbol{K}\bar{\boldsymbol{e}}[k])$ can be written as Relation (3.18).

$$\boldsymbol{\Delta u}[k] = -[\boldsymbol{I}\,\boldsymbol{0}\ldots\boldsymbol{0}]\boldsymbol{\Gamma}^{-1}\boldsymbol{\gamma} \tag{3.16}$$

$$= \underbrace{-[\boldsymbol{I}\,\boldsymbol{0}\ldots\boldsymbol{0}](\boldsymbol{P}^T_{\bar{\Delta u}}\boldsymbol{Q}\boldsymbol{P}_{\bar{\Delta u}} + \boldsymbol{R}^{-1})\boldsymbol{P}_{\bar{\Delta u}}\boldsymbol{Q}}_{\boldsymbol{K}}\,\tilde{\bar{\boldsymbol{e}}}[k] \tag{3.17}$$

$$\boldsymbol{\Delta u}[k] = \boldsymbol{K}\tilde{\bar{\boldsymbol{e}}}[k] \tag{3.18}$$

Unconstrained Basic MPC for LPV Systems

To improve the readability of this section, the parameter dependencies in ω are not always explicitly stated.

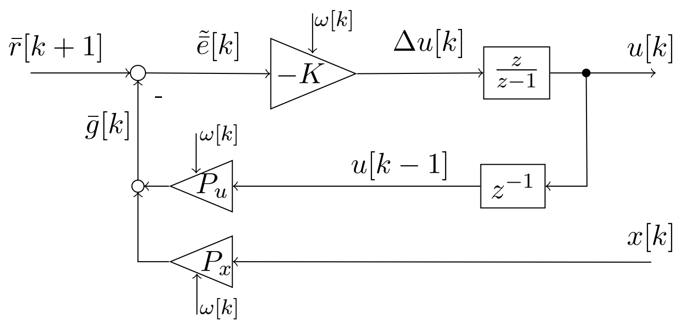


Figure 3.7: Unconstrained Basic MPC structure for LPV systems

Inserting $\tilde{\bar{\boldsymbol{e}}}[k] = \bar{\boldsymbol{r}}[k+1] - \bar{\boldsymbol{g}}[k]$, according to the structure depicted of Figure 3.7, the Control Law (3.18) can be expanded to

$$\bar{\boldsymbol{\Delta u}}[k] = \boldsymbol{K}\left(\bar{\boldsymbol{r}}[k+1] - (\boldsymbol{P}_u\boldsymbol{u}[k-1] + \boldsymbol{P}_x\boldsymbol{x}[k])\right)$$

With the output accumulation $\boldsymbol{u}[k] = \boldsymbol{u}[k-1] + \bar{\boldsymbol{\Delta u}}[k]$ one derives:

$$\boldsymbol{u}[k] = \boldsymbol{u}[k-1] - \boldsymbol{K}\boldsymbol{P}_u\boldsymbol{u}[k-1] + \boldsymbol{K}\bar{\boldsymbol{r}}[k+1] - \boldsymbol{K}\boldsymbol{P}_x\boldsymbol{x}[k]$$

For that system, the measured states and the reference prediction vector act as input

$\boldsymbol{v}_{C,b}[k] = \boldsymbol{x}[k]\ \bar{\boldsymbol{r}}[k+1]]^{\mathrm{T}}$ to the controller and the stator voltage as output $\boldsymbol{y}_{C,b}[k] = \boldsymbol{u}[k]$.

$$\boldsymbol{u}[k] = \underbrace{\Big[\boldsymbol{I} - \boldsymbol{K}\boldsymbol{P}_u(\omega[k])\Big]}_{\boldsymbol{A}_{C,b}} \underbrace{\boldsymbol{u}[k-1]}_{\boldsymbol{\xi}_{C,b}[k-1]} \quad + \quad \underbrace{\Big[\boldsymbol{K} \quad -\boldsymbol{K}\boldsymbol{P}_x(\omega[k])\Big]}_{\boldsymbol{B}_{C,b}} \underbrace{\begin{bmatrix} \bar{\boldsymbol{r}}[k+1] \\ \boldsymbol{x}[k] \end{bmatrix}}_{\boldsymbol{v}_{C,b}[k-1]} \tag{3.19}$$

$$\boldsymbol{y}_{C,b}[k] = \underbrace{\boldsymbol{A}_{C,b}}_{\boldsymbol{C}_{C,b}} \boldsymbol{\xi}_{C,b}[k-1] + \underbrace{\boldsymbol{B}_{C,b}}_{\boldsymbol{D}_{C,b}} \boldsymbol{v}_{C,b}[k-1] = \boldsymbol{u}[k]. \tag{3.20}$$

The spectrum of the unconstrained Basic Controller's system matrix $\boldsymbol{A}_{C,b}(\omega)$, was quantitatively analyzed by plotting its eigenvalues for discrete values of $\omega \in \mathcal{W} = [-10^3, 10^3]$ rad/s. The result can be seen in Figure 3.8.

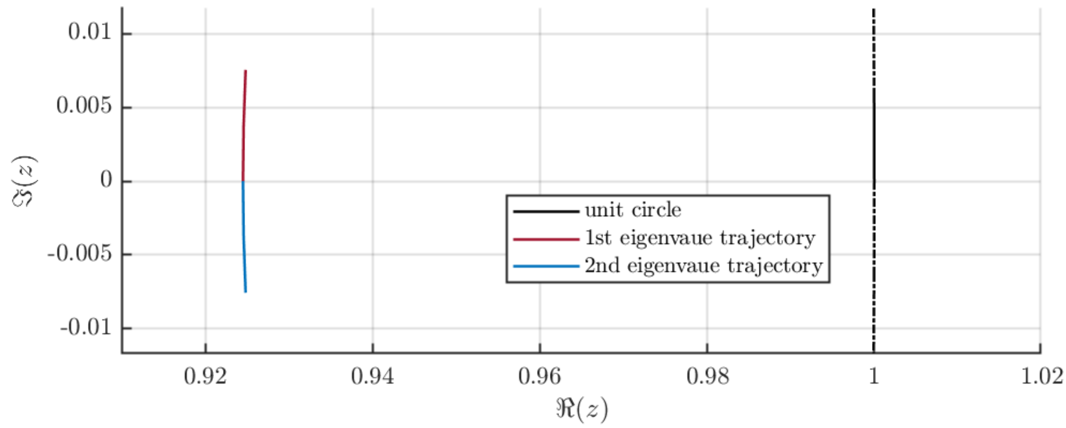


Figure 3.8: Spectrum of $\boldsymbol{A}_{C,b}(\omega)$ for $\omega \in \mathcal{W}$ of the unconstrained Basic MPC.

By deriving the transfer function matrix $\boldsymbol{v} \rightarrow \boldsymbol{y}$ for frozen parameter $\omega = \omega_l$, the frequency response can be analyzed.

$$\boldsymbol{G}_{vy}\big|_{\omega_l} = \boldsymbol{C}_{C,b}\big|_{\omega_l} \left(z\boldsymbol{I} - \boldsymbol{A}_{C,b}\big|_{\omega_l}\right)^{-1} \boldsymbol{B}_{C,b} + \boldsymbol{D}_{C,b}$$

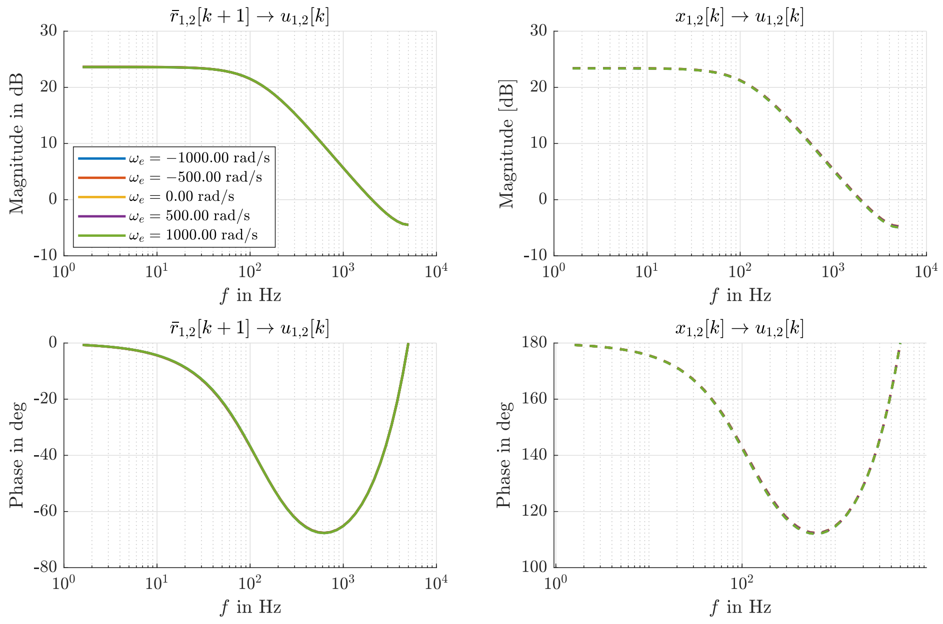


Figure 3.9: Discrete-time frequency response for the unconstrained Basic MPC.

The different frozen transfer functions are fairly similar, with their magnitude response differing in the sub-dB range. Thus, they appear to be nearly identical in Figure 3.9.

Unconstrained Hybrid MPC for LPV Systems

Again, to improve the readability of this section, the parameter dependencies ω are not explicitly stated.
The reconstructed plant input is defined by Equation (3.21).

$$\tilde{\boldsymbol{u}}[k-1] = \boldsymbol{B}^{-1}\left(\boldsymbol{x}[k] - \boldsymbol{A}\boldsymbol{x}[k-1]\right) \tag{3.21}$$

The control law is linear in terms of the quasi-tracking error with the optimal control gain $\boldsymbol{K}$ according to Equation (3.18) $\Delta\boldsymbol{u}[k] = \boldsymbol{K}\tilde{\boldsymbol{e}}[k]$. The accumulated plant input is then given by

$$\boldsymbol{u}[k] = \boldsymbol{u}[k-1] + \boldsymbol{K}\tilde{\boldsymbol{e}}[k] \tag{3.22}$$

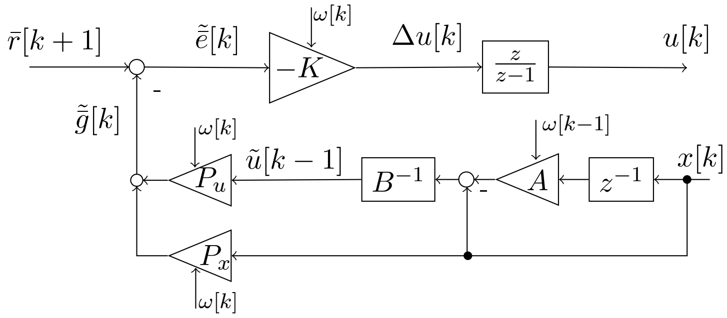


Figure 3.10: Unconstrained Hybrid MPC structure for LPV systems

Using $\tilde{\boldsymbol{e}}[k] = \bar{\boldsymbol{r}}[k+1] - \tilde{\bar{\boldsymbol{g}}}[k]$, with the definition of $\tilde{\bar{\boldsymbol{g}}}[k]$ according to the structure depicted in Figure 3.10, Equation (3.22) can be expanded to:

$$\begin{aligned}\boldsymbol{u}[k] &= \boldsymbol{u}[k-1] + \boldsymbol{K}\left(\bar{\boldsymbol{r}}[k+1] - \boldsymbol{P}_u\left(\boldsymbol{B}^{-1}\left(\boldsymbol{x}[k] - \boldsymbol{A}\boldsymbol{x}[k-1]\right)\right) - \boldsymbol{P}_x\boldsymbol{x}[k]\right)\\ &= \boldsymbol{u}[k-1] + \boldsymbol{K}\boldsymbol{P}_u\boldsymbol{B}^{-1}\boldsymbol{x}[k-1] + \boldsymbol{K}\bar{\boldsymbol{r}}[k+1] - \boldsymbol{K}\boldsymbol{P}_u\boldsymbol{B}^{-1}\boldsymbol{x}[k] - \boldsymbol{K}\boldsymbol{P}_x\boldsymbol{x}[k].\end{aligned}$$

After introducing a shift register for $\boldsymbol{x}[k-1]$, this can be written in matrix notation

$$\begin{bmatrix}\boldsymbol{u}[k]\\ \boldsymbol{x}[k]\end{bmatrix} = \begin{bmatrix}\boldsymbol{I} & \boldsymbol{K}\boldsymbol{P}_u\boldsymbol{B}^{-1}\boldsymbol{A}\\ \boldsymbol{0} & \boldsymbol{0}\end{bmatrix}\begin{bmatrix}\boldsymbol{u}[k-1]\\ \boldsymbol{x}[k-1]\end{bmatrix} + \begin{bmatrix}-\boldsymbol{K} & \boldsymbol{K}\left(\boldsymbol{P}_u\boldsymbol{B}^{-1} + \boldsymbol{P}_x\right)\\ \boldsymbol{0} & \boldsymbol{I}\end{bmatrix}\begin{bmatrix}\bar{\boldsymbol{r}}[k+1]\\ \boldsymbol{x}[k]\end{bmatrix} \tag{3.23}$$

For that system, the measured states and the reference $\boldsymbol{v}_{C,h} = [\bar{\boldsymbol{r}}\ \boldsymbol{x}]^{\mathrm{T}}$ act as input to the controller and the stator voltage as output $\boldsymbol{y}_{C,h}[k] = \boldsymbol{u}[k]$. To derive a state space formulation the state vector is defined as the following $\boldsymbol{\xi}^T_{\boldsymbol{C,h}} = [\boldsymbol{u}\ \boldsymbol{x}]$.

$$\begin{bmatrix}\boldsymbol{u}[k]\\ \boldsymbol{x}[k]\end{bmatrix} = \underbrace{\begin{bmatrix}\boldsymbol{I} & -\boldsymbol{K}\boldsymbol{P}_u(\omega[k])\boldsymbol{B}^{-1}\boldsymbol{A}(\omega[k-1])\\ \boldsymbol{0} & \boldsymbol{0}\end{bmatrix}}_{\boldsymbol{A}_{C,h}}\begin{bmatrix}\boldsymbol{u}[k-1]\\ \boldsymbol{x}[k-1]\end{bmatrix} + \tag{3.24}$$

$$\underbrace{\begin{bmatrix}-\boldsymbol{K} & \boldsymbol{K}\left(\boldsymbol{P}_u(\omega[k])\boldsymbol{B}^{-1} + \boldsymbol{P}_x(\omega[k])\right)\\ \boldsymbol{0} & \boldsymbol{I}\end{bmatrix}}_{\boldsymbol{B}_{C,h}}\begin{bmatrix}\bar{\boldsymbol{r}}[k+1]\\ \boldsymbol{x}[k]\end{bmatrix} \tag{3.25}$$

$$\boldsymbol{y}_{C,h}[k] = [\boldsymbol{I}\ \boldsymbol{0}]\cdot\boldsymbol{A}_{C,h}\cdot\boldsymbol{\xi}[k-1] + [\boldsymbol{I}\ \boldsymbol{0}]\cdot\boldsymbol{B}_{C,h}\cdot\boldsymbol{v}_{C,h}[k-1] = \boldsymbol{u}[k] \tag{3.26}$$

According to the LPV State Space Model (3.26), the system matrix $\boldsymbol{A}_{C,h}$ is an upper block-triangular matrix. Thus the eigenvalues are on the main diagonal of it. Since those entries are constant, the spectrum of this system matrix is given by

$$\sigma\left(\boldsymbol{A}_{C,h}\left(\omega\right)\right) = \{1, 1, 0, 0\}$$

With two discrete-time eigenvalues at one, it can be said, that this controller is of intergrating behavior.
Again, the transfer function matrix $\boldsymbol{v} \to \boldsymbol{y}$ for frozen parameter $\omega = \omega_l$ is derived, so that the frequency response can be analyzed.

$$\boldsymbol{G}_{vy}\big|_{\omega_l} = \boldsymbol{C}_{C,h}\big|_{\omega_l}\left(z\boldsymbol{I} - \boldsymbol{A}_{C,h}\big|_{\omega_l}\right)^{-1}\boldsymbol{B}_{C,h} + \boldsymbol{D}_{C,h}$$

The obtained frequency responses for $\omega \in [-1 \ \ -0.5 \ 0 \ 0.5 \ 1] \cdot 10^3$ rad/s are depicted in Figure 3.11.

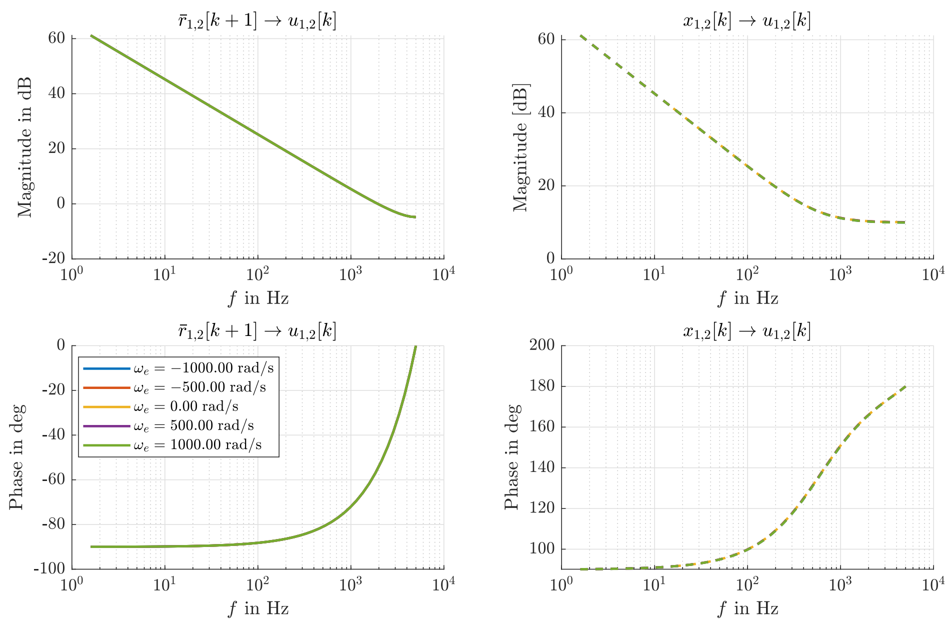


Figure 3.11: Discrete-time frequency response for the unconstrained Hybrid MPC.

The different frozen transfer functions are fairly similar, with their magnitude response differing in the sub-dB range. Thus, they appear to be nearly identical in Figure 3.11.

Unconstrained MPC with Embedded Integrator for LPV Systems

If the notation used in this thesis is applied to Equation (1.29) of Wang's Book [12] for the optimal control signal of the linear unconstrained augmented system $\Sigma_\Delta : (\boldsymbol{A}_\Delta, \boldsymbol{B}_\Delta, \boldsymbol{C}_\Delta,)$

(3.14), the result is given by Equation (3.27). Where $\{\mathcal{P}_{\Delta u}, \mathcal{P}_{\Delta x}\}$ are the prediction matrices for the state feedback and input of the augmented system Σ_Δ, similar to the structure of $\{\boldsymbol{P}_{\bar{\Delta u}}, \boldsymbol{P}_x\}$ as given by Equation (3.5).

$$\bar{\boldsymbol{\Delta u}}[k]^* = [\boldsymbol{I}\ \boldsymbol{0}\ldots\boldsymbol{0}]\left(\mathcal{P}_{\Delta u}^{\mathrm{T}}\mathcal{P}_{\Delta u} + \boldsymbol{R}\right)^{-1}\left(\mathcal{P}_{\Delta u}^{\mathrm{T}}\mathcal{P}_r\boldsymbol{r}[k+1] - \mathcal{P}_{\Delta u}^{\mathrm{T}}\mathcal{P}_{\Delta x}\underbrace{[\Delta\boldsymbol{x}[k]\ \boldsymbol{y}[k]]}_{\boldsymbol{\xi}_\Delta[k]}\right) \tag{3.27}$$

$$= \boldsymbol{K}_y\left(\boldsymbol{r}[k+1] - \boldsymbol{y}[k]\right) + \boldsymbol{K}_x\Delta\boldsymbol{x}[k] \tag{3.28}$$

By doing a coefficient comparison with of Equations (3.27) and (3.28), the linear control gains can be determined. With that the Diagram 3.12 for the locally unconstrained controller can be derived.

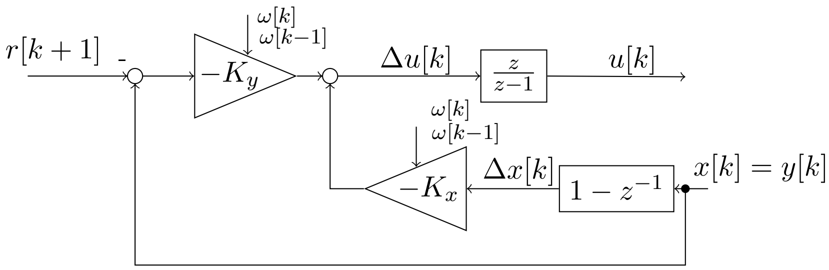


Figure 3.12: Unconstrained MPC with embedded integrator accoding to Wang [12] for LPV systems using the notation of this thesis.

Using the same method for the derivation of the state space model for the Basic and Hybrid MPC, the dynamics for the unconstrained embedded integrator MPC formulation is derived.

$$\begin{bmatrix}\boldsymbol{u}[k]\\ \boldsymbol{x}[k]\end{bmatrix} = \underbrace{\begin{bmatrix}\boldsymbol{I} & \boldsymbol{K}_x\\ \boldsymbol{0} & \boldsymbol{0}\end{bmatrix}}_{\boldsymbol{A}_{C,ei}}\begin{bmatrix}\boldsymbol{u}[k-1]\\ \boldsymbol{x}[k-1]\end{bmatrix} + \underbrace{\begin{bmatrix}-\boldsymbol{K}_y & \boldsymbol{K}_y - \boldsymbol{K}_x\\ \boldsymbol{0} & \boldsymbol{I}\end{bmatrix}}_{\boldsymbol{B}_{C,ei}}\begin{bmatrix}\boldsymbol{r}[k+1]\\ \boldsymbol{x}[k]\end{bmatrix} \tag{3.29}$$

Again, the controller's system matrix $\boldsymbol{A}_{C,ei}$ of Equation (3.29) an upper block-triangular. Thus, its spectrum can be easily written down:

$$\sigma\left(\boldsymbol{A}_{C,ei}\left(\omega\right)\right) = \{1, 1, 0, 0\}.$$

With that the integrating behavior in the unconstrained operation can be directly observed.

3.4.2 Comparison of the MPC Formulations with inactive Constraints

The Basic MPC stores the last control signal $\boldsymbol{u}[k-1]$ and uses it directly for the prediction. This results directly in feedback structure of the accumulator $z/(z-1)$ as seen in Figure 3.7. As confirmed by the eigenvalue and transfer function analysis this results in low-pass behavior. Therefore this approach does not guarantee offset-free tracking under constant disturbances such as model faults. Section 3.5.2 introduces an extension of this formulation for integral behavior. However, as it will be shown, this extension requires anti-windup measures for implementation on real hardware.
Using an augmented incremental state space system $\Sigma_\Delta : (\boldsymbol{A}_\Delta, \boldsymbol{B}_\Delta, \boldsymbol{C}_\Delta,)$ according to Wang [12] the accumulator is in direct series connection to the input of the plant. However, the augmented state space system Σ_Δ with the augmented state vector $\boldsymbol{\xi}_\Delta[k] = [\Delta\boldsymbol{x}[k]\ \boldsymbol{y}[k]]^{\mathrm{T}}$ hinders the flexibility of this formulation for extensions, such as the ones presented in the following Section.
To ensure integral behavior and use absolute states $\boldsymbol{x}$ the direct feedback of the accumulator was interrupted by estimating the previous control signal $\tilde{\boldsymbol{u}}[k-1] = \boldsymbol{B}^{-1}\left(\boldsymbol{x}[k] - \boldsymbol{A}\boldsymbol{x}[k-1]\right)$ which can be thought of as the actual last control signal disturbed by an input disturbance $\tilde{\boldsymbol{u}}[k-1] = \boldsymbol{u}[k-1] + \boldsymbol{d}_1[k-1]$. This formulation is untypical, since it requires $\boldsymbol{B}^{-1}$ which is in general not guaranteed to exist, but poses no issues for the model of the PMSM (3.3). As seen by the transfer functions of Figure 3.11 and the spectrum of the controller dynamic matrix, this leads to integral action. With that, offset free tracking under constant disturbances is ensured.
If the MPC operates with active constraints, the high frequency dynamics might be worth studying, to gain insight if one of those three presented approaches tends to less robust behavior with respect to model and timing uncertainties. For that the Bode Plots 3.9 and 3.9 are consulted. Looking at high frequencies within the sampling limit $5\,\mathrm{kHz} \leq f < T_S^{-1}/2$, the magnitude attenuation of the Hybrid formulation is close to identical to the Basic formulation for $\boldsymbol{r} \rightarrow \boldsymbol{u}$, but the Basic MPC exhibits 15 dB more attenuation for all frozen transfer functions of state feedback to actuating signal $\boldsymbol{x} \rightarrow \boldsymbol{u}$ than the Hybrid MPC.

3.5 Extensions

Three extensions for the model-predictive field oriented control of PMSM where chosen. First the issue of computation and switching delay is addressed with deadtime compensation. Section 3.5.1 describes the well-established approach for one-step compensation. Chapter 4 explores methods to compensate for fractional delays.
Since the Basic MPC formulation according to [7] does not compensate for e.g. model faults, an extension is presented which solves this by considering the integral tracking error in the cost function.
Finally, to allow for short-term violations of current limits, soft constraints are introduced.

3.5.1 Deadtime Compensation

For clarity, the prediction notation $u[i|j]$ is introduced, where $i, j \in \mathbb{Z}$. Here, i denotes the time of application or measurement of the variable, and j denotes the time of information acquisition or computation. Thus, for example, $u[k+1|k]$ is the control variable calculated at time k, which takes effect starting at $k+1$.

Due to the one-step input delay of the computation of the PWM's duty cycle, the control variable $u[k+1|k]$ calculated at time k does not influence the transition from k to $k+1$, but rather the transition from $k+1$ to $k+2$. Therefore, the MPC prediction does not begin with $\hat{y}[k+1|k]$, but with $\hat{y}[k+2|k]$. The starting point is the known intermediate prediction based on $\Sigma_k : [\boldsymbol{A}(\omega[k]), \boldsymbol{B}, \boldsymbol{E}(\omega[k])]$.

$$\hat{\boldsymbol{x}}[k+1|k] = \boldsymbol{A}(\omega[k])\boldsymbol{x}[k|k] + \boldsymbol{B}\boldsymbol{u}[k|k-1] + \boldsymbol{E}(\omega[k]) \tag{3.30}$$

where $\boldsymbol{u}[k|k-1]$ is the control variable calculated in the previous sampling step and now in effect. The optimization variable thus starts at $\Delta\boldsymbol{u}[k+1|k]$, and the first predicted output influenced by it is $\hat{\boldsymbol{y}}[k+2|k]$.

Delay compensation affects both the control variable calculation and the transformation from the rotor-fixed to the stator-fixed coordinate system $dq \to \alpha\beta$. Due to the equidistant, PWM-synchronous sampling, the pre-calculated angle

$$\hat{\varphi}_e[k+1|k] = \varphi_e[k|k] + T_S\omega_e[k|k]$$

is used instead for the transformations between SRF and RRF.

At time k, because $\boldsymbol{C} = \boldsymbol{I}$, the output $\boldsymbol{y}[k|k] = \boldsymbol{x}[k|k]$ is measured. Based on this, the

MPC calculates the output sequence

$$\{\hat{\boldsymbol{y}}[k+2|k], \ldots, \hat{\boldsymbol{y}}[k+N_P|k]\}$$

for the control input sequence

$$\{\Delta\boldsymbol{u}[k+1|k], \ldots, \Delta\boldsymbol{u}[k+N_C|k]\}$$

and determines $\boldsymbol{u}[k+1|k]$ from this by minimizing the cost function (3.7). The measurement result $\boldsymbol{y}[k+1|k+1]$ is still determined by the control input $\boldsymbol{u}[k|k-1]$ that is already active; the newly calculated control input $\boldsymbol{u}[k+1|k]$ takes effect only in the following sampling interval.

Implementation

Without deadtime compensation, the objective of the quadratic program is to determine the prediction vector $\bar{\boldsymbol{\Delta u}}[k]$. This vector is composed of:

$$\bar{\boldsymbol{\Delta u}}^*[k] = \begin{bmatrix} \boldsymbol{\Delta u}^*[k] \\ \boldsymbol{\Delta u}^*[k+1] \\ \vdots \\ \boldsymbol{\Delta u}^*[k+N_C-1] \end{bmatrix} = \begin{bmatrix} \Delta u_d^*[k] \\ \Delta u_q^*[k] \\ \Delta u_d^*[k+1] \\ \Delta u_q^*[k+1] \\ \vdots \\ \Delta u_d^*[k+N_C-1] \\ \Delta u_q^*[k+N_C-1] \end{bmatrix}$$

To compensate the deadtime, the optimization target must be

$$\bar{\boldsymbol{\Delta u}}^*[k+1|k] = \begin{bmatrix} \boldsymbol{\Delta u}^*[k+1|k] \\ \vdots \\ \boldsymbol{\Delta u}^*[k+N_C-1|k] \end{bmatrix}$$

.

with that, the new cost function is

$$J_{\mathrm{dtc}} = \|\bar{\boldsymbol{y}}[k+2] - \bar{\boldsymbol{r}}[k+2]\|_{\boldsymbol{Q}}^2 + \left\|\bar{\boldsymbol{\Delta u}}[k+1|k]\right\|_{\boldsymbol{R}}^2 \tag{3.31}$$

To achieve this, the prediction equation (3.5) must be modified according to:

$$\bar{\boldsymbol{y}}[k+2|k] = \underbrace{\tilde{\boldsymbol{P}}_x \hat{\boldsymbol{x}}[k+1|k] + \tilde{\boldsymbol{P}}_u \boldsymbol{u}[k|k-1] + \tilde{\boldsymbol{P}}_E \bar{\boldsymbol{E}}(\hat{\omega}[k+1|k])}_{=:\bar{\boldsymbol{g}}[k+1|k]} + \tilde{\boldsymbol{P}}_{\bar{\Delta u}} \bar{\boldsymbol{\Delta u}}[k+1|k] \tag{3.32}$$

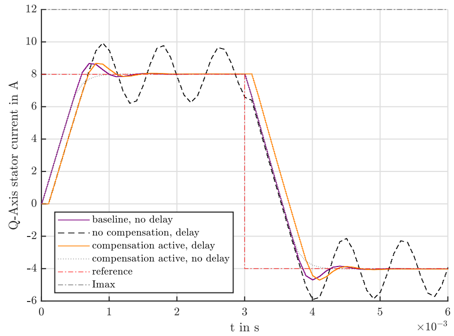
Or in more detail:

$$\begin{aligned}
\underbrace{\begin{bmatrix} \hat{\boldsymbol{y}}[k+2] \\ \vdots \\ \hat{\boldsymbol{y}}[k+D_s] \end{bmatrix}}_{\bar{\boldsymbol{y}}[k+2|k]} &= \underbrace{\begin{bmatrix} \boldsymbol{C}\left(\boldsymbol{A}(\omega[k])\right)^2 \\ \vdots \\ \boldsymbol{C}\left(\boldsymbol{A}(\omega[k])\right)^{D_s} \end{bmatrix}}_{\tilde{\boldsymbol{P}}_x} \bar{\boldsymbol{x}}[k] + \underbrace{\begin{bmatrix} \boldsymbol{C}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{B} \\ \vdots \\ \boldsymbol{C}\left(\sum_{l=0}^{D_s-1} \left(\boldsymbol{A}(\omega[k])\right)^l\right)\boldsymbol{B} \end{bmatrix}}_{\tilde{\boldsymbol{P}}_u} \boldsymbol{u}[k|k-1] \\
&+ \underbrace{\begin{bmatrix} \boldsymbol{C}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{B} & \boldsymbol{C}\boldsymbol{B} & \cdots & \boldsymbol{0}_2 \\ \boldsymbol{C}\left(\left(\boldsymbol{A}(\omega[k])\right)^2 + \boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{B} & \boldsymbol{C}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right)\boldsymbol{B} & \ddots & \vdots \\ \vdots & \vdots & \ddots & \boldsymbol{0}_2 \\ \boldsymbol{C}\left(\sum_{l=0}^{D_s-1} \left(\boldsymbol{A}(\omega[k])\right)^l\right)\boldsymbol{B} & \boldsymbol{C}\left(\sum_{l=0}^{D_s-2} \left(\boldsymbol{A}(\omega[k])\right)^l\right)\boldsymbol{B} & \cdots & \boldsymbol{C}\boldsymbol{B} \end{bmatrix}}_{\tilde{\boldsymbol{P}}_{\bar{\Delta u}}} \underbrace{\begin{bmatrix} \boldsymbol{\Delta u}[k+1] \\ \vdots \\ \boldsymbol{\Delta u}[k+D_s-1] \end{bmatrix}}_{\bar{\boldsymbol{\Delta u}}[k+1|k]} \\
&+ \underbrace{\begin{bmatrix} \boldsymbol{C}\left(\boldsymbol{A}(\omega[k]) + \boldsymbol{I}\right) \\ \vdots \\ \boldsymbol{C}\left(\sum_{l=0}^{D_s-1} \left(\boldsymbol{A}(\omega[k])\right)^l\right) \end{bmatrix}}_{\tilde{\boldsymbol{P}}_E} \boldsymbol{E}(\omega[k]).
\end{aligned} \tag{3.33}$$

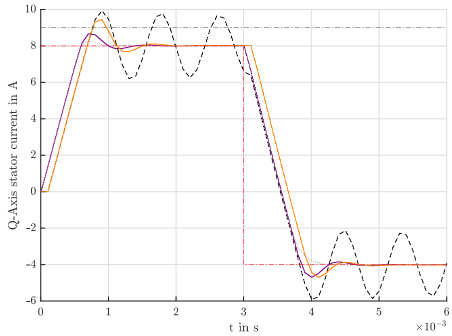
Where D_s are the number of decision steps and is given by $D_S = N_P - 1$.

Comparison across the three MPC Formulations

For the simulation, the PMSM Model (2.3) with parameters of the NW3 machine at the electric machine laboratory of TUG (see Table A.1) was simulated in a locked rotor scenario at $\omega_m = 0$ rad/s. For this simulation, an ideal pulse-amplitude modulation, synchronous to the controller sampling rate of 100 µs, is employed and the MPC is built on a perfect model of the machine. The tuning parameters used for this simulation, can be obtained from Table A.2.

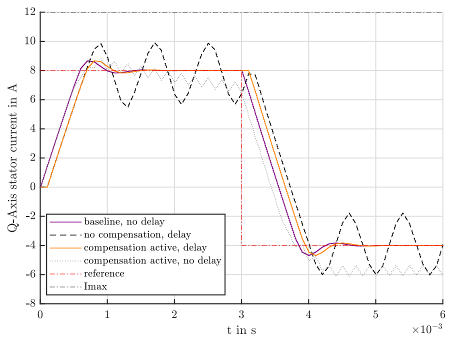


(a) Soft constraint $I_{max} = 12\,\text{A}$

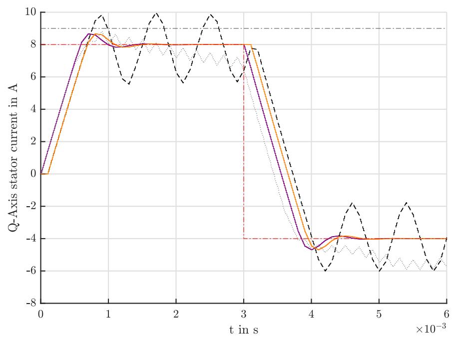


(b) Soft constraint $I_{max} = 9\,\text{A}$

Figure 3.13: Simulated step responses for deadtime scenarios with the Basic MPC

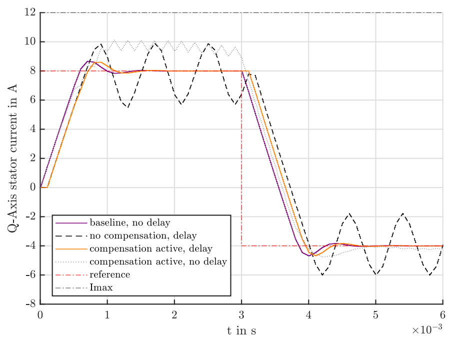


(a) Soft constraint $I_{max} = 12\,\text{A}$

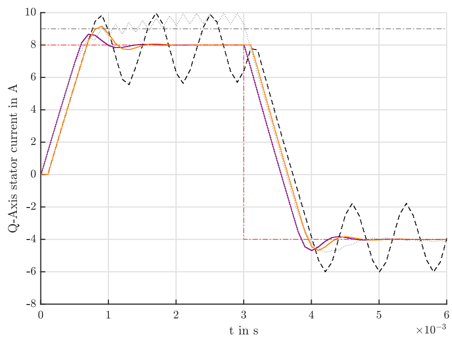


(b) Soft constraint $I_{max} = 9\,\text{A}$

Figure 3.14: Simulated step responses for deadtime scenarios with the Embedded Integrator MPC due to Wang



(a) Soft constraint $I_{max} = 12\,\text{A}$



(b) Soft constraint $I_{max} = 9\,\text{A}$

Figure 3.15: Simulated step responses for deadtime scenarios with the Hybrid MPC

The voltage limit was set to $U_{DC} = 50\sqrt{3}$ V, in order to test constraint operations as well. For this parameter set, the voltage constraint will be active for approximately 5 µs after positive reference step and 7 µs after the negative step. Each of those simulation is also conducted with active and inactive current soft constraints, to check the effect of the deadtime compensation on that as well. The results are shown in Figures 3.13 3.14 and 3.15.

3.5.2 Integral State Augmentation for the Basic MPC

Diese Herangehensweise ist in der LV Optimization and Control vorgestellt worden
Ich hab sie aber in den dort referenzierten "großen" MPC-Büchern [7] [3] nicht finden können
Habe diesbzüglich e-mail ans IRT geschickt

The main idea is to calculate the accumulated tracking error with a designated dynamical system Σ_z for each sampling step.

$$\Sigma_z : \boldsymbol{z}[k+1] = \boldsymbol{z}[k] + (\boldsymbol{y}[k] - \boldsymbol{r}[k])$$

Based on this system, the prediction for $\boldsymbol{z}[k+1]$ over D_S is calculated. This prediction vector is then used as an additional quadratic term, weighted by an diagonal matrix $\boldsymbol{W}_I$ in the cost function, to minimize the integral error.

$$J_{is} = \cdots + \|\bar{\boldsymbol{z}}\|^2_{\boldsymbol{W_I}} \tag{3.34}$$

To directly use this in the Quadratic Program (3.6), the prediction vector $\bar{\boldsymbol{z}}[k+1]$ is expressed in terms of $\bar{\boldsymbol{\Delta u}}[k]$. Using a similar procedure as described in Section 3.1.3, the prediction vector for $\boldsymbol{z}$ is calculated by the recursive predictions

$$\begin{aligned}
\boldsymbol{z}[k+1] &= \boldsymbol{z}[k] + \boldsymbol{y}[k] - \boldsymbol{r}[k], \\
\boldsymbol{z}[k+2] &= \boldsymbol{z}[k] + (\boldsymbol{y}[k] - \boldsymbol{r}[k]) + (\boldsymbol{y}[k+1] - \boldsymbol{r}[k+1])\,, \\
&\vdots \\
\boldsymbol{z}[k+N_P] &= \boldsymbol{z}[k] + \sum_{l=0}^{N_P-1} (\boldsymbol{y}[k+l] - \boldsymbol{r}[k+l])\,.
\end{aligned}$$

. Using matrix notation this can be written as

$$\bar{\boldsymbol{z}}[k+1] = (\boldsymbol{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{z}[k] + \boldsymbol{P}_T\,(\bar{\boldsymbol{y}}[k] - \bar{\boldsymbol{r}}[k+1])$$

where P_T is lower triangular block matrix consisting of identity matrices as elements. Substituting the Output Prediction (3.5),

$$\bar{\boldsymbol{y}}[k+1] = \bar{\boldsymbol{g}}[k] + \boldsymbol{P}_{\Delta u}\bar{\boldsymbol{\Delta u}}[k]$$

into the recursive predictions above, the stacked integral-state prediction becomes

$$\begin{aligned}\bar{\boldsymbol{z}}[k+1] &= (\mathbf{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{z}[k] + \boldsymbol{P}_T\left(\bar{\boldsymbol{g}}[k] + \boldsymbol{P}_{\Delta u}\bar{\boldsymbol{\Delta u}}[k] - \bar{\boldsymbol{r}}[k+1]\right) \\ &= \underbrace{(\mathbf{1}_{N_P} \otimes \boldsymbol{I}_2)\boldsymbol{z}[k] + \boldsymbol{P}_T\left(\bar{\boldsymbol{g}}[k] - \bar{\boldsymbol{r}}[k+1]\right)}_{:=\boldsymbol{\rho}_I,\ \text{known at instance } k} + \underbrace{\boldsymbol{P}_T\boldsymbol{P}_{\Delta u}}_{:=\boldsymbol{P}_I,\ \text{related to } \bar{\boldsymbol{\Delta u}}[k]} \bar{\boldsymbol{\Delta u}}[k].\end{aligned}$$

Due to the separation of the term $\boldsymbol{\rho}_I$ and the $\boldsymbol{P}_I$, $\bar{\boldsymbol{z}}[k+1] = \boldsymbol{\rho}_I + \boldsymbol{P}_I\bar{\boldsymbol{\Delta u}}[k]$, can easily be inserted into the modfied cost function J_{is} of Equation (3.34).

$$J_{is} = \|\bar{\boldsymbol{y}} - \bar{\boldsymbol{r}}\|^2_{\boldsymbol{Q}} + \|\bar{\boldsymbol{\Delta u}}\|^2_{\boldsymbol{R}} + \|\boldsymbol{\rho}_I + \boldsymbol{P}_I\bar{\boldsymbol{\Delta u}}[k]\|^2_{\boldsymbol{W}_I}$$

Defining the auxiliary terms

$$\begin{aligned}\boldsymbol{Q}_I &= \boldsymbol{P}_I^{\mathrm{T}}\boldsymbol{W}_I\boldsymbol{P}_I \\ \boldsymbol{\gamma}_I^{\mathrm{T}} &= \boldsymbol{\gamma}^{\mathrm{T}} + 2\boldsymbol{\rho}_I^{\mathrm{T}}\boldsymbol{W}_I\boldsymbol{P}_I \\ \boldsymbol{\Gamma}_I &= \boldsymbol{P}_{\bar{\Delta u}}^{\mathrm{T}}(\boldsymbol{Q} + \boldsymbol{Q}_I)\boldsymbol{P}_{\bar{\Delta u}} + \boldsymbol{R}\end{aligned}$$

eases the augmentation such that the quadratic program with integral states can be expressed as

$$\begin{aligned}&\arg\min_{\bar{\boldsymbol{\Delta u}}_S} \ \frac{1}{2}\|\bar{\boldsymbol{\Delta u}}[k]\|^2_{\boldsymbol{\Gamma}_I} + \tilde{\gamma}_I{}^{\mathrm{T}}\bar{\boldsymbol{\Delta u}}[k] \\ &\text{s.t.} \quad \boldsymbol{M}\bar{\boldsymbol{\Delta u}}[k] \leq \boldsymbol{\mu}\end{aligned} \tag{3.35}$$

The weighting matrix $\boldsymbol{W}_I$ is a tune-able factor. Figure 3.16 illustrates the influence of $\boldsymbol{W}_I$ on the step response for $\mathbf{0} \to \boldsymbol{i}_{S,ref} = [-2, 30]^{\mathrm{T}}$ A, considering a mismatch between the parameters of the MPC and the actual machine. For that simulation, an ideal inverter was used, $\omega_m = 0$ rad/s and the simulated model mismatch of the MPC is with respect to the motor parameters of Table A.1. The pulse-amplitude modulation is synchronous to the controller sampling rate of 100 µs,

Remarks

Using this approach, integral action of the MPC can be ensured, without increasing the elements of optimization variables or sacrificing the Basic MPC formulation. However, if the MPC operates under active constraints and the tracking error cannot be fully compensated, the implicit integral error state z will wind up. Looking at the cost function,

$$J_{is} = \|\bar{\boldsymbol{y}} - \bar{\boldsymbol{r}}\|^2_{\boldsymbol{Q}} + \|\bar{\boldsymbol{\Delta u}}\|^2_{\boldsymbol{R}} + \|\boldsymbol{z}\|^2_{\boldsymbol{W}_I}$$

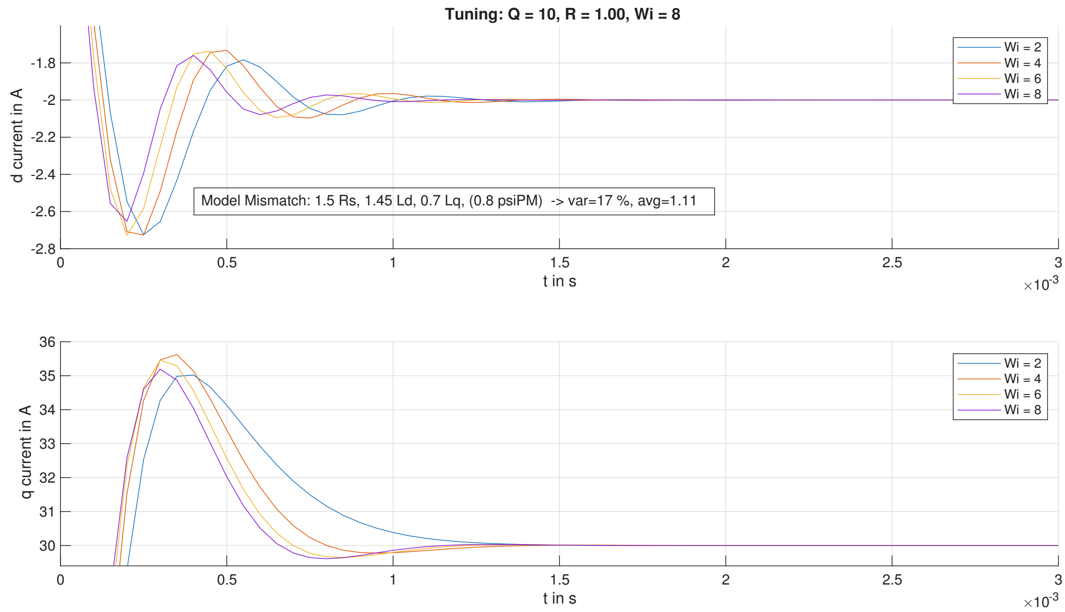


Figure 3.16: Comparison of the step responses for multiple values of W_I considering model fault

one can see that the quadratic form regarding z will also grow in value.The solver of the quadratic program will still search for the vector $\bar{\boldsymbol{\Delta u}}[k]$ which minimizes that cost. Even though the effect of the weighting $\boldsymbol{Q}, \boldsymbol{R}$ will still be considered, by that, numerical issues may arise due to that wind up effect.

3.5.3 Backup Stage

If

In classical PI-FOC with SVPWM,

Two common methods are Minimum Phase Error and Minimum Amplitude Error over-modulation strategies. For implementation details, refer to [9]

3.5.4 Soft Constraints

By adding a slackness variable η_i to the left hand side of a linear inequality according to Equation 3.36

$$m_i \zeta_i \leq \mu_i + \eta_i \tag{3.36}$$

where m_i, μ_i are scalar constants, the optimization variable ζ_i can violate this constraint. For every step in time, the constraint slackness $m_i \zeta_i[k] - \mu_i = \eta_i[k]$ is then minimized by an extra term in the cost function with a weighting factor w_η. This results in a

mechanism, which allows for short-term constraint violations, due to the modification of Equation (3.36), but the linear term ηw_η of J, minimizes the slackness.
The vector of optimization variables is extended by the slackness variable $\bar{\boldsymbol{\Delta u}} \rightarrow \boldsymbol{\zeta} = \begin{bmatrix} \bar{\boldsymbol{\Delta u}}^{\mathrm{T}} & \eta \end{bmatrix}^{\mathrm{T}}$. To reduce the number of elements in $\boldsymbol{\zeta}$, it was decided to use one scalar slackness variable, which is scaled by the constant vector $\boldsymbol{\rho} \,:\, \rho_i \in \mathbb{R},\ 0 \leq \rho_i \leq 1$. The linear inequalities of the Quadratic Program (3.6) are extended to

$$\boldsymbol{M\zeta} \leq \boldsymbol{\mu} + \boldsymbol{\rho}\eta$$

. Considering the slackness variable and integral error states, the total cost function is then defined by

$$\tilde{J}_{\text{tot}} = \|\bar{\boldsymbol{y}} - \bar{\boldsymbol{r}}\|^2_{\boldsymbol{Q}} + \|\bar{\boldsymbol{\Delta u}}\|^2_{\boldsymbol{R}} + \|\bar{\boldsymbol{z}}\|^2_{\boldsymbol{W_I}} + w_\eta \eta$$

.

With that, the total modified Quadratic Program (3.37) can be written down.

$$\begin{aligned} \bar{\boldsymbol{\Delta u}}^*[k] = \underbrace{\begin{bmatrix} 1 & 0 & & 0 \\ 0 & \ddots & & \vdots \\ & 0 & 1 & 0 \end{bmatrix}}_{\left[\boldsymbol{I}^{mD_S \times mD_S} \,\vdots\, \boldsymbol{0}^{mD_S \times 1}\right]} & \quad \arg\min_{\boldsymbol{\zeta}} \boldsymbol{\zeta}^T \begin{bmatrix} \tilde{\boldsymbol{\Gamma}} & \boldsymbol{0} \\ \boldsymbol{0}^T & w_\eta \end{bmatrix} \boldsymbol{\zeta} + \begin{bmatrix} \gamma^T & 0 \end{bmatrix} \boldsymbol{\zeta} \\ & \text{s.t.} \quad \begin{bmatrix} \boldsymbol{M} & -\boldsymbol{\rho} \\ \boldsymbol{0}^T & 0 \end{bmatrix} \underbrace{\begin{bmatrix} \bar{\boldsymbol{\Delta u}} \\ \eta \end{bmatrix}}_{\boldsymbol{\zeta}} \leq \begin{bmatrix} \boldsymbol{\mu} \\ / \end{bmatrix} \end{aligned} \tag{3.37}$$

The matrix $\left[\boldsymbol{I}^{mD_S \times mD_S} \,\vdots\, \boldsymbol{0}^{mD_S \times 1}\right]$ is used in order for $\bar{\boldsymbol{\Delta u}}[k]$ to be the optimization target (without η). In the implementation $[\boldsymbol{I}_2\ \boldsymbol{0}_2\ \ldots\ \boldsymbol{0}_2\ 0]$ instead of $\left[\boldsymbol{I} \,\vdots\, \boldsymbol{0}\right]$, such that only the first voltage vector $\boldsymbol{\Delta u}[k]$ is applied.

3.6 Chosen Implementation

3.6.1 Controller Structure

3.6.2 Simulation

To validate the Hybrid MPC Implementation with deadtime compensation and soft constraints, a speed profile was simulated using MATLAB® Simulink®. In this simulation,

the MPC has an imperfect model of the PMSM (for PMSM parameters, see Table A.1 and tuning parameters Table A.2), with $1.25 \cdot R_S, 1.12 \cdot L_d, 1.17 \cdot L_q, 0.97 \cdot \psi_{PM}$, an ideal pulse-amplitude modulation inverter with one-step switching delay and the mechanical system, modeled as a damped oscillator, according to Equation (2.5).
For this simulation the PMSM was controlled in speed mode with the model predictive current controller and an PI speed controller (with back-calculation for anti-windup). Figure 3.17 depicts the speed reference and the resulting reference currents due to the speed controller.

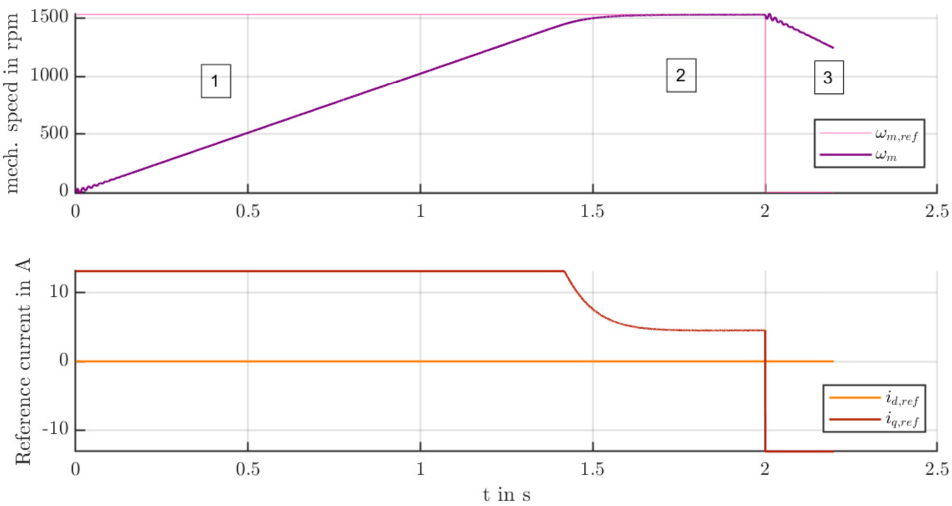


Figure 3.17: Simulated speed ramp: reference quantities

As it can be seen in Figure 3.17 and 3.18, three different phases were distinguished:

1. **Acceleration**: To accelerate the machine from standstill, the speed controller requests torque using the maximum current. Due to the differential speed on the machine shaft, an oscillation can be observed.

2. **Over-Modulation**: After approximately 1.4 s, the required stator voltage space vector reaches the limit of the inscribed circle at $U_{DC}/\sqrt{3}$. The voltage constrained of the current controller were artificially lowered, in order to test the constrained operation. With $U_{DC} = 140\sqrt{3}$ and a speed reference of $n_{ref,1} = 1530\,\text{rpm}$, the voltage hexagon will be fully utilized.

3. **Deceleration**: At $t = 2\,\text{s}$, the speed controller initiates braking by requesting negative current in the Q axis. Due to the change in speed, the mechanical system will oscillate.

The quantities of Figure 3.17 can be interpreted as inout quantities for MPCC, which outputs the stator voltage in order to control the stator according to the Optimization Problem (3.37). The resulting stator voltages and currents are depicted in Figure 3.18. To visualize the hexagonal current constraints, the limits regarding inscribed circle at $U_{DC}/\sqrt{3}$ and the outer circle $2/3 \cdot U_{DC}$ are depicted as well as the circular current constraint $\|\boldsymbol{i}_S\| \leq I_{max}$.

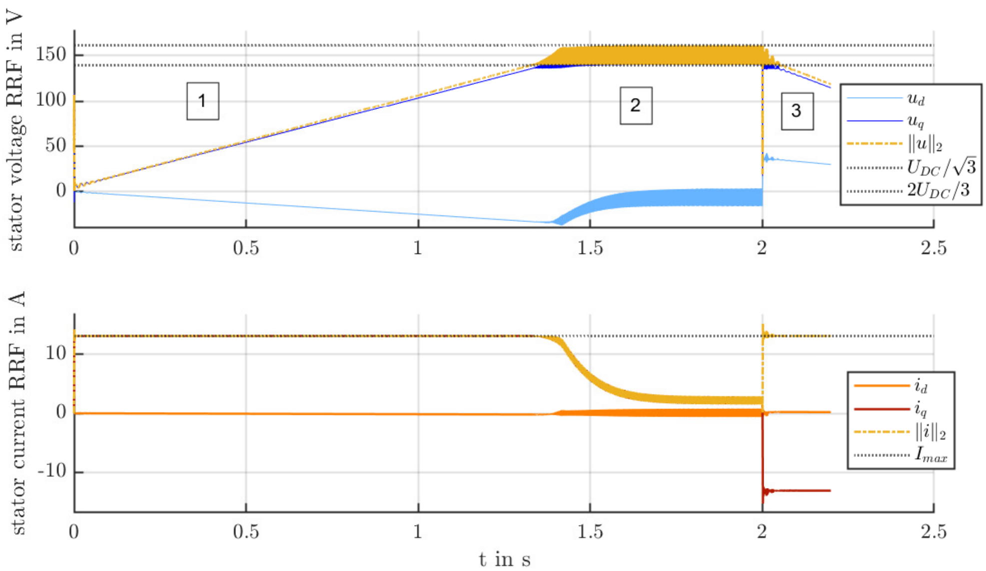


Figure 3.18: Simulated speed ramp: control quantities

In Figure 3.18 one can observe the oscillations in voltage and current during the overmodulation phase. In the following, these quantities are depicted in detail for each of the three phases separately (see Figures 3.21, 3.22 and 3.23).

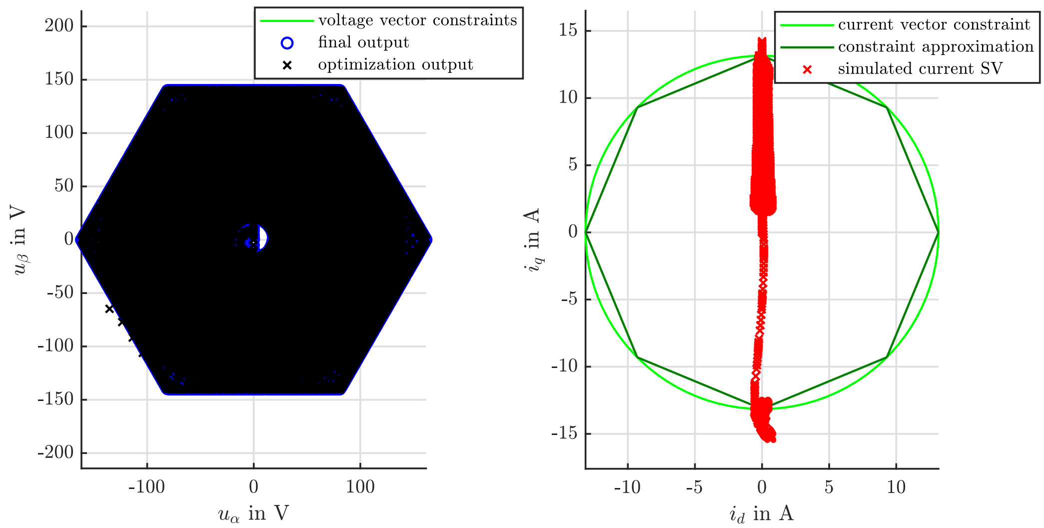


Figure 3.19: Simulated speed ramp: voltage and current constraints

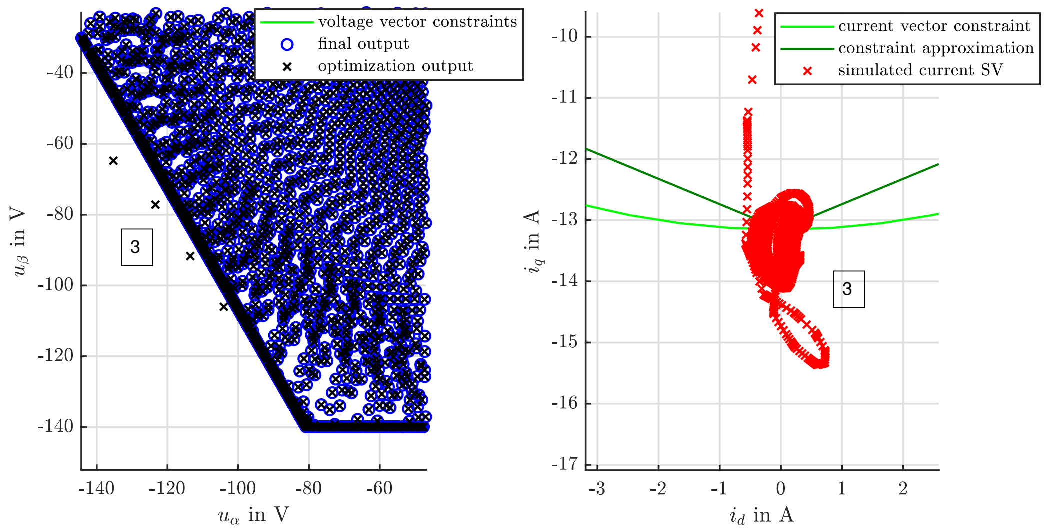
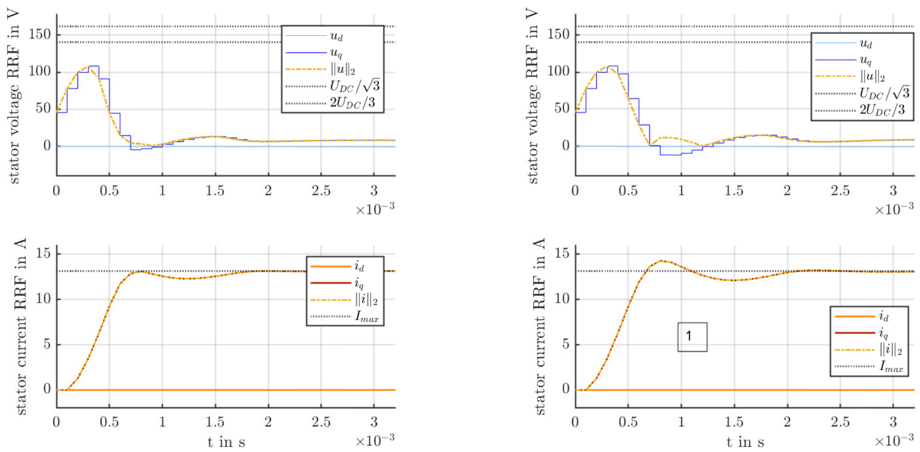


Figure 3.20: Simulated speed ramp: voltage and current constraints, detail during deceleration phase

As it can be seen the output of the MPC, would violate the voltage constraints by up to 5V during the deceleration phase. The first outlier happens 7 µs and the last 11 µs after the reference change.
This issue does not improve if the MPC is provided with an actual reference trajectory, instead of copying the value for the current reference over D_s. Since the rotor speed deceleration is at maximum during that interval, it can be assumed that the prediction

error due to $\hat{\omega}[k+1] = \omega[k]$ is to blame for that.
In the following the stator voltages and currents for the acceleration, constant maximum speed and deceleration phase will be discussed.



(a) Without soft constraints (b) With soft constraints

Figure 3.21: Simulated speed ramp: Phase 1 - Acceleration

In Figure 3.21 the initial step response for the current controller is depicted for the case of hard and soft current constraints. With the use of soft constraints, one can achieve a faster convergence rate. However, since the controller parameters were not adapted in that simulation, this benefit of soft constraints cannot be observed here.

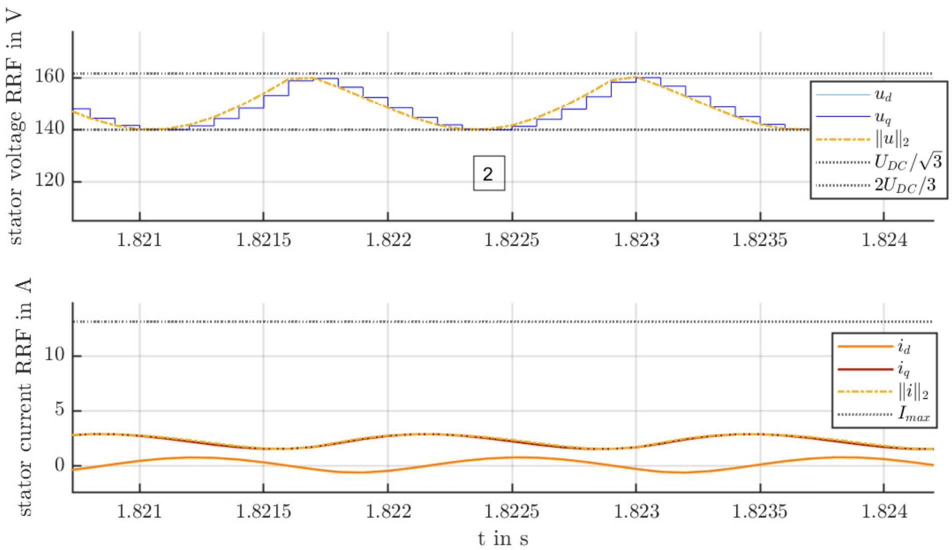


Figure 3.22: Simulated speed ramp: Phase 2 - Over-modulation due to active voltage constraints

Figure 3.22 depicts the stator quantities during continuous maximum speed. Since the $\boldsymbol{u}_S$ travels at the border of the voltage hexagon, one can see oscillations in its waveform with peak-to-peak amplitude of $(2/3 - 1/\sqrt{3})U_{DC}$ and an angular frequency of $6\omega_e$. The D axis current also exhibits an ripple, but does not contribute much to the length of the voltage space vector due to its small amplitude. Due to those oscillations, the current and electric torque will also show a ripple.

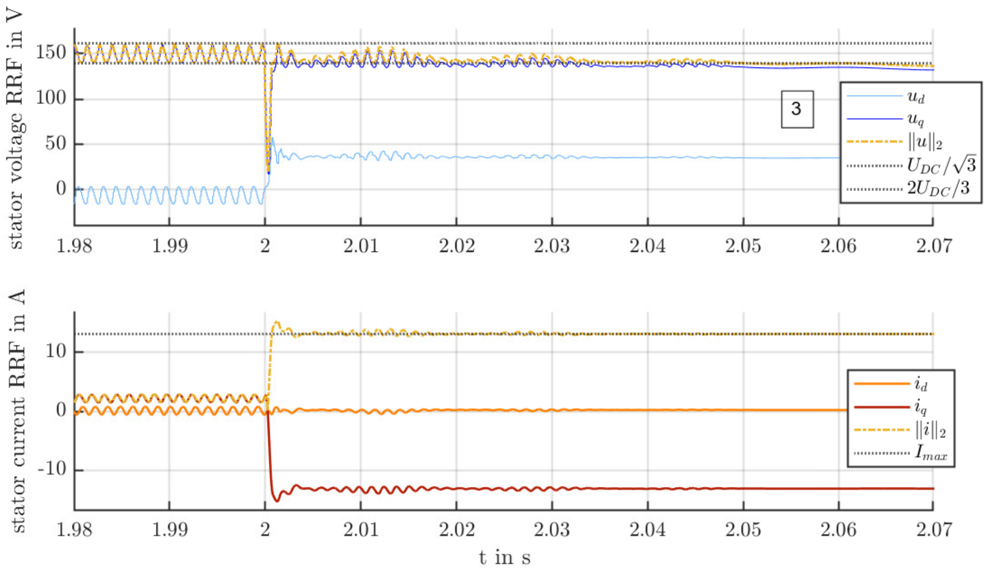


Figure 3.23: Simulated speed ramp: Phase 3 - Deceleration

In the last phase, shown in Figure 3.23, the deceleration phase can be inspected. At $t = 2\,\mathrm{s}$, the electric brake with $i_{q,ref} < 0$ is initiated. Due to the sudden change in rotor speed, the mechanical system will start to oscillate. The resulting ripple in ω_e enters into the controller as scheduling variable. Due to the assumption of constant rotor speed over the prediction horizon this may result in a further model mismatch. TODO: schauen ob da Γ immer noch positiv definit

3.7 Experimental Evaluation

3.7.1 Test Setup

The experimental evaluation is conducted on a motor test bench located at the Electric Drives and Machines Institute of Graz University of Technology. The setup consists of an induction machine and a PMSM (see Table A.1), which are mechanically coupled.
Each machine is supplied by a dedicated 2 Level VSI, with the electrical interconnections

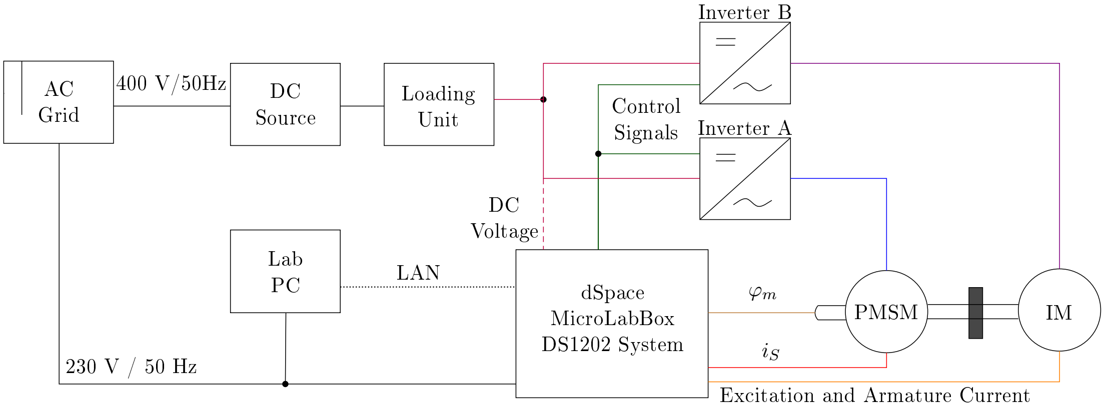
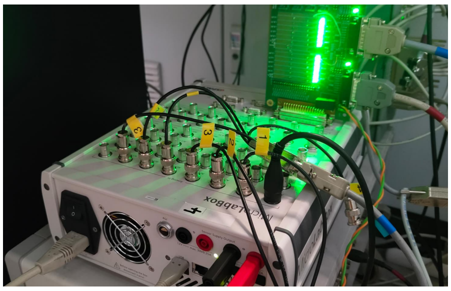
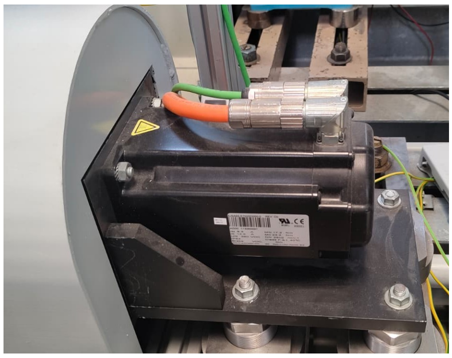


Figure 3.24: Schematic of the end-to-end test bed for the MPC tests at TUG



(a) dSpace MicroLabBox



(b) PMSM

Figure 3.25: Photos of the test bed at TUG

of the individual drives shown in Figures 3.24 and ??. The rotor position is measured by a position sensor mounted on the shaft of the PMSM with an incremental encoder.

The control algorithms are implemented in MATLAB® using user-defined MATLAB functions, while the simulation models are developed and evaluated in MATLAB® Simulink®. Real-time execution of the control algorithms is carried out using a dSPACE DS1202 MicroLabBox. The interaction with the real-time control system and the adjustment of control parameters are performed on a host PC using ControlDesk by dSPACE.

3.7.2 Step Response

3.7.3 Constrained Operation

3.8 Comparison of the MPC Approaches

3.8.1 Steady State Behavior

3.8.2 Timing Uncertainties

3.8.3 Constrained Operation

3.8.4 Computational Effort

3.8.5 Extension: time optimal pre-rotation

In [16]

4 Observer based Current Prediction

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A
Appendix

A.1 Machine Parameters

Table A.1: Parameters of the NW3 PMSM

Quantity	Symbol	Value
Machine type	–	8JSA65.E7025D0-0 Rev. C0
Serial number	–	AD00 113065997
Number of pole pairs	p	5
Nominal current	I_N	9.3 A
Nominal torque	M_N	17.2 Nm
Maximum current	I_max	12.5 A
Nominal DC-link voltage	U_N	560 VDC
Nominal speed	n_N	2500 rpm
Stator resistance	R_s	0.41 Ω
Stator inductance	$L_d = L_q$	3.5 mH
Permanent magnet flux linkage	ψ_{PM}	0.1843 Vs

Table A.2: Nominal tuning parameter values for the model-predictive current controller

Quantity	Symbol	Value
Number of decision steps	D_s	2
Tracking weight	Q	$2 \cdot 10^3$
Control increment weight	R	1
Cost-based Integral Tracking error weight	W_I	5
Soft constraint weight	w_η	$1 \cdot 10^4$

Note: $\boldsymbol{Q} \in \mathbb{R}^{qD_s \times qD_s}$, $\boldsymbol{R}, \boldsymbol{W}_I \in \mathbb{R}^{mD_s \times mD_s}$ are diagonal weighting matrices with identical diagonal elements.